

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/386601242>

Optimizing an integrated hybrid energy system with hydrogen-based storage to develop an off-grid green community for sustainable development in Bangladesh

Article in *International Journal of Hydrogen Energy* · January 2025

DOI: 10.1016/j.ijhydene.2024.11.454

CITATIONS

3

4 authors, including:



[Asif Jaman](#)

Rajshahi University of Engineering and Technology

5 PUBLICATIONS 9 CITATIONS

[SEE PROFILE](#)

READS

171

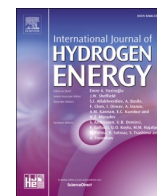


[Barun Kumar Das](#)

Edith Cowan University

103 PUBLICATIONS 3,688 CITATIONS

[SEE PROFILE](#)



Optimizing an integrated hybrid energy system with hydrogen-based storage to develop an off-grid green community for sustainable development in Bangladesh[☆]

Asif Jaman^{a,1}, Rafid Al Mahmud^{a,1}, Barun K. Das^{b,*}, **Mohammad Shahed H.K. Tushar^a**

^a Department of Mechanical Engineering, Rajshahi University of Engineering & Technology, Rajshahi, 6204, Bangladesh

^b School of Engineering, Edith Cowan University, Joondalup, 6027, Australia

ARTICLE INFO

Keywords:

Hybrid system
Reliability
Human development
Environmental emissions
Cost of energy

ABSTRACT

An integrated renewable system that utilizes solid waste-based biogas is important steps towards the sustainable energy solutions to rural off-grid communities in Bangladesh. In this study, a hybrid energy system consisting of photovoltaic modules, wind turbines, biogas generators, fuel cells, and electrolyzer-hydrogen tank-based energy storage is optimized using non-dominated sorting genetic algorithm (NSGA-II). The hybrid system is optimized based on the cost of energy and human health damage as objective functions, and a fuzzy decision-making technique is employed to determine the optimal solution to the multi-objective approach. Additionally, several economic, ecological, and social indicators are also investigated while meeting a certain load reliability. An energy management strategy has been developed in the MATLAB environment to satisfy the community load and the battery-driven electric vehicle load. Results from this comprehensive analysis suggest that the optimal configuration of PV/WT/FC/BG has an energy cost of 0.1634 \$/kWh, and an ecosystem damage of 0.00098 species/year. The human health damage and the human development index of the optimized system are 0.1732 DALYs and 0.696 DALYs, respectively. Additionally, the proposed system has a lifecycle emission of 123,730 kg CO₂-eq/year, carbon emission penalties of \$1856/year, a job creation potential of 30 jobs/MW over the 25 years of project lifetime. The hybrid system oversees solid waste management solutions and provides the community with sustainable energy and vehicle recharge.

1. Introduction

As of 2024, Bangladesh ranks as the 10th most densely populated country, and its pursuit of nationwide electrification persists amid increasing economic expansion [1]. The combustion of fossil fuels for electricity generation has been traditional; nonetheless, the consequent emission of hazardous gasses and environmental deterioration has prompted concern among environmentalists and the policy makers [2, 3]. Electricity generation in Bangladesh has been highly reliant on natural gas, heavy fuel oil (HFO), diesel fuel, and coal, where natural gas and coal contribute to about 69% of total power generation [4]. However, the reserves of gas and coal are limited. The world is increasingly transitioning to alternative energy sources such as solar, wind, and tidal to generate clean energy [5,6]. As one of the largest and 4th most

populous cities in Bangladesh, Rajshahi is continuously faced with the dilemma of meeting community load demand [1]. Additionally, there has been a major increase in battery-operated vehicles (BEVs) in the streets of Rajshahi [7]. These vehicles are solely rely on the main grid for their main power source, affecting the community demand side and causing grid overload [8]. This has a negative impact on both power quality and grid stability, prompting the authority to interrupt the continuous flow of electricity at the mass level. A plausible solution is to install hybrid renewable energy sources (HRES) and to develop energy management strategies that will result in unremitting energy and subsequently eliminate the adversity of the intermittent nature of such energy sources [9]. Likewise, relying on such a hybrid system improves reliability and results in low greenhouse gas emissions compared to its single renewable energy source-based counterparts [10].

[☆] A full research paper submitted for publication in “International Journal of Hydrogen Energy”.

* Corresponding author.

E-mail address: b.das@ecu.edu.au (B.K. Das).

¹ These authors contributed equally.

In retrospect, the study area, Rajshahi has an enormous solar potential [11]. However, due to the intermittent nature of the renewables sources, the continuous power generation to satisfy the demand is not guaranteed [12–15]. Consequently, to tackle this erratic behavior, the most conventional solution is to install a controlled power source [16, 17]. In this context, a biogas generator in this system will work as the controlled power source to serve as backup and improve the reliability of the HRES. Additionally, a proton-exchange membrane fuel cell (PEMFC) can be implemented in HRES in place of a battery due to its nature of continuously generating electricity, given there is a supply of oxygen and hydrogen. This is especially important as when dealing with the fluctuating load demand of the EV charging station. Hydrogen storage provides a sustainable energy storage solution because of its high energy density, low rate of self-discharge, and wide scalability [18, 19]. Additionally, hydrogen functions as an energy carrier, hence enabling its storage and transportation for future utilization [20–22].

Several studies have been carried out to develop and optimize HRES technologies while ensuring a stable and cost-effective power supply to different communities. However, most of the literature used HRES systems only to supply electrical, hydrogen, and thermal loads based on the techno-economic and environmental indicators. In this context, Abdin et al. [23] carried out a feasibility analysis of a hydrogen-based HRES for both off-grid and on-grid configurations and found that the electricity costs of 0.32 \$/kWh and 0.034 \$/kWh for the off-grid and on-grid, respectively. Rathod and Balaji [24] optimized a PV/WT/DG/BATT/FC-based HRES for meeting an electric load demand using improved red tailed hawk algorithm and found the cost of energy (COE) of 0.3772 \$/kWh, NPC of \$ 44,05,900, and the annual cost of \$ 3, 10,000. Modu et al. [25] employed the levy flight-salp swarm algorithms to optimize PV/WT/BG/BATT/FC-based HRES and reported the COE of 0.933162 \$/kWh. Ahmed et al. [26] used NSGA-II to evaluate the optimal sizing of the PV/WT/DG/BATT/ROD with methanation process for fulfilling the electric, water, and cooking gas demand for a remote island loads. The study conducted by Yan et al. [27] assessed the effects of various design factors on a micro-grid based on PV/WT/FC. The analysis revealed a minimum COE of 0.3458 \$/kWh and an annual total cost of \$11,952. The benefits of introducing hydrogen storage to the stand-alone PV-based system have been reported by Tatti et al. [28] and found that the higher load reliability of 99% can be achieved by hydrogen storage than the battery storage (93%). Wahedi et al. [29] demonstrated the feasibility of a novel HRES-based fast electric charging station in Qatar, consisting of PV, wind turbine, electrolyzer, and biogas generator, and found the minimum COE to be 0.285 \$/kWh. However, the study was based on a single objective, and it did not consider socio-environmental factors. Cuesta et al. [30] conducted a study showing how HRES-based systems can contribute to social benefit factors such as job creation or the human development index (HDI). The framework provided can help the HRES designers and planners assess the eligibility of the systems and optimize them in terms of multiple social benefit indicators. Padrón et al. [31] analyzed the feasibility of a PV/WT/BG system for an autonomous desalination system considering the technical and economic factors using the HOMER Pro software. Mayer et al. [32] explored the pros and cons of a single objective function-based optimization over its multi-objective counterparts using genetic algorithm for a PV/WT/BAT system and found out that multi-objective optimization in a conflicting scenario gives a more vivid picture of a set of solutions. In order to comprehend the decarbonization of the transportation sector and the potential contribution to the global net zero goal, Daramola et al. [33] looked at the economic and environmental benefits of integrating battery storage systems into renewable sources, combined heat and power systems, and electric vehicles. When battery systems were integrated, the study indicated that both the operational cost and CO₂ emissions decreased by 47.2%. The system was grid-connected and natural gas-fueled fuel cell was utilized in the system. Despite its design to power electric vehicles and meet thermal demands; environmental and sustainable indicators were overlooked. A

PV/WT/Geothermal-based power and hydrogen generation system is developed by Yilmaz et al. [34]. Sizing of PV/WT/PEM electrolyzer-based grid-connected hydrogen production system is investigated by Fabianek and Madlener [35] based on techno-economic indicators. The summary of recent works on the HRES is reported in Table 1.

Based on the literature review, it is appeared that the research works on the investigation of a hybrid renewable energy system that integrates photovoltaic (PV), wind turbine (WT), and biogas generator technologies, coupled with a fuel cell-electrolyzer-based storage facility to satisfy the community and the EVCS loads are limited. This research introduces a hybrid energy system combining solar PV, wind turbine, biogas generator, and fuel-cell electrolyzer-based storage facility, tailored for densely populated areas like Katakali, Rajshahi. The system uniquely supports both community power needs and nearby EV charging stations through an efficient dual load management. The study employs a combination of optimization criteria—human health impact and energy cost—alongside additional indicators such as job creation, ecological benefits, and carbon penalties. The objective functions, performance indicators, and the methodological approach to achieve those parameters are presented in Fig. 1. This research work also explores how varying load demands influence these performance metrics, highlighting the system's innovation and broader societal benefits. Key contributions of this study are reported as follows:

- An integrated microgrid system including fuel cell-electrolyzer-based storage facilities, solar PV modules, wind turbines, and biogas generators is thoroughly assessed based on key socio-environmental and economic factors. Three different HRESs including PV/WT/FC/BG, PV/WT/BG, PV/WT/FC have been compared in this regard based on techno-economic, human health damage, job creation, and ecological indicators.
- Biogas potential has been assessed using the MSW data indigenous to the study area and based on that, biogas generators are added to the hybrid system. The integration of biogas into the HRES is sure to have a significant positive impact to an energy starved country like Bangladesh.
- To meet the needs for community and EVCS power, an extensive structure and energy management plan have been established, with the majority of the power demand being met by using the extra energy produced by the hybrid energy system.
- An untried combination of objective functions has been considered in this study, human health damage (HHD) and cost of energy (COE) for the purpose of system optimization. Energy management strategies have been development in MATLAB environment and the HRES is optimized using the multi-objective NSGA-II algorithm. A fuzzy based multi-criteria decision-making technique is used to determine the final solution on the Pareto front. Additionally, several indicators including job creations, ecological benefits, carbon penalty have been reported.
- The impacts of incremental load demand on the performance indicators have been discussed.

In an effort to lessen the consequences of climate change and improve energy security, the main goal of this research is to promote the adoption of clean and sustainable energy alternatives. The methodological approach developed and the findings reported in this study may be very helpful in creating environmentally friendly and robust energy systems for executing such projects in similar climatic areas such as Rajshahi city, Bangladesh.

2. Materials and methods

2.1. Geographical information and renewable resources of study area

The study area (Fig. 2) for this research is centered on the Rajshahi

Table 1

Summarized literature review of recent works on HRES.

HRES model	Software/ Algorithms utilized	Study area	Consumption nature	Performance parameters		Findings and limitations	Year	Ref
				COE (\$/kWh)	NPC (\$)			
PV/WT/FC/ BAT	HOMER	Oman	Electrical, hydrogen	COE: 0.0158 \$/kWh COH: 0.401 \$/kg	529,361	Hybrid system is designed to meet electrical and hydrogen load for transportation sector using HOMER software tool. Although economic benefits are exclusively examined, the environmental and ecological benefits are not reported.	2024	[36]
PV/FC/ TEG/TIG/ Grid	HOMER	Pakistan	Electrical, hydrogen, thermal	COE: 0.1084 \$/kWh COH: 6.94 \$/kg	25.95 M	This study examines the grid-connected and standalone HRES system for electric, thermal, and hydrogen loads based on techno-economics and environmental considerations. However, the job creations and social benefits are examined.	2024	[37]
PV/WT/ Hydro/ FC/GT/ MGT/PTC	TRNSYS	Italy	Electric, EV	0.1464	–	TRNSYS software tool is used to design an HRES to satisfy electric and vehicle loads based on economic analysis only.	2024	[38]
PV/WT/ BAT/ Grid/PEM	ParetoSearch optimization algorithm	USA	Hydrogen, EV	COE: 0.0858–0.1258 \$/kWh COH: 6–13 \$/kg	–	The research develops predictive models for hydrogen generation and its utilization in hydrogen-powered vehicles. Optimization of the HRES is conducted using MATLAB-based ParetoSearch, focusing solely on techno-economic indicators.	2024	[39]
PV/WT/FC/ Grid	HOMER	China	Electric, hydrogen	0.283	8351442	HOMER software tool is used to size both grid connected and stand-alone HRES for meeting electric and hydrogen loads for refueling stations based on the economic parameters only.	2024	[40]
PV/WT/FC/ BAT/Grid	HOMER	Kuwait	Electric, hydrogen	COE: 0.539 \$/kWh COH: 6.85 \$/kg	9.91 M	Economic analysis is examined using HOMER software for a grid connected and stand-alone HRES for electric and hydrogen loads. However, ecological, and social benefits have not been discussed.	2024	[41]
PV/WT/BAT	HOMER	Morocco	Electrical, EV	0.0841	2.99 M	Supermarket and EV loads are satisfied by the grid connected HRES system, which is optimized by HOMER based on the economic and environmental consideration of the grid electricity. The LCE of the renewable components are not considered in their study.	2023	[42]
PV/WT/BG/ BAT/ Flywheel	HOMER	Jordan	Electrical	0.024	298,259	Biogas-based solar, wind, battery, and flywheel HRES with both grid connected and off-grid options are optimized using HOMER software tool. Although details sensitivity analysis along with techno-economic-environmental parameters are analyzed, social benefits are not reported.	2023	[43]
PV/WT/GT/ BAT	NSGA-II	China	Electrical	0.0567	–	Large scale HRES including H ₂ fueled gas turbine is optimized using multi-objective NSGA-II lifecycle analysis. However, the final solution is not reached by using multi-criteria decision-making process.	2023	[44]
PV/WT/DG/ BAT	HOMER, MATLAB, JAYA	USA	EV	0.09–0.34	111720–339240	HOMER software with MATLAB interface is used to develop a demand response program for meeting the EV load based on the techno-economic indicators. Although detailed economic and sensitivity have been reported, ecological and job creation are not discussed.	2023	[45]
PV/FC/BAT	HOMER	India	Electric, hydrogen	COE: 0.41–0.48 \$/kWh COH: 3–3.22 \$/kg	–	Electric and hydrogen for fuel cell-based EV loads are satisfied by the PV-based system. Energy and hydrogen costs, and emissions by replacing fossil fuel vehicles have been discussed. However, the LCE and ecological benefits are not reported.	2022	[46]
PV/WT/BG/ FC	NSGA-II	India	Electric, water	COE: 0.1967 \$/kWh COW: 0.86 \$/m ³	–	HRES for meeting electric and freshwater demands is optimized using NSGA-II based on the economic, and human health damage. The job creation and ecological benefits have also been assessed. However, the detailed solid waste-based	2022	[47]

(continued on next page)

Table 1 (continued)

HRES model	Software/ Algorithms utilized	Study area	Consumption nature	Performance parameters		Findings and limitations	Year	Ref
				COE (\$/kWh)	NPC (\$)			
PV/WT/BAT	HOMER	Greece	Electrical	0.08 €/kWh	–	biogas resource assessment and the EV load demand with the VAWT wind turbine have not considered. Grid-connected with and without net metering options have been modelled using HOMER software tool using three different battery technologies. GHG emissions along with economic parameters have been examined; however, the job creation and hydrogen storage-based system are not considered.	2021	[48]
PV/WT/ BAT/TLC	HOMER, GA	Australia	Electrical, heating	0.226	1,243,383	HRES is optimized by HOMER and compare with GA while meeting the electric and thermal loads. EV loads, and social and ecological benefits are not considered.	2021	[49]
PV/PHS/ MGT-CHP	HOMER, MATLAB	Iran	Electrical, thermal	0.09	42.5 M	HOMER and MATLAB are used to optimized while meeting the electric and thermal loads. Detailed techno-economic analyses are reported; however, human development index, job creations and social benefits are not examined.	2021	[50]
PV/WT/BG/ FC	NSGA-II	Bangladesh	Electric, EV	COE, NPC, HHD, HDI, JC, carbon penalty, sensitivity of load increment		Multi-objective optimization using NSGA-II in the MATLAB platform is used to determine the optimal sizing of the HRES components based on cost of energy and human health damage as objective functions. A fuzzy decision-making process is applied to determine the optimal solution on the Pareto front.	2024	Present study

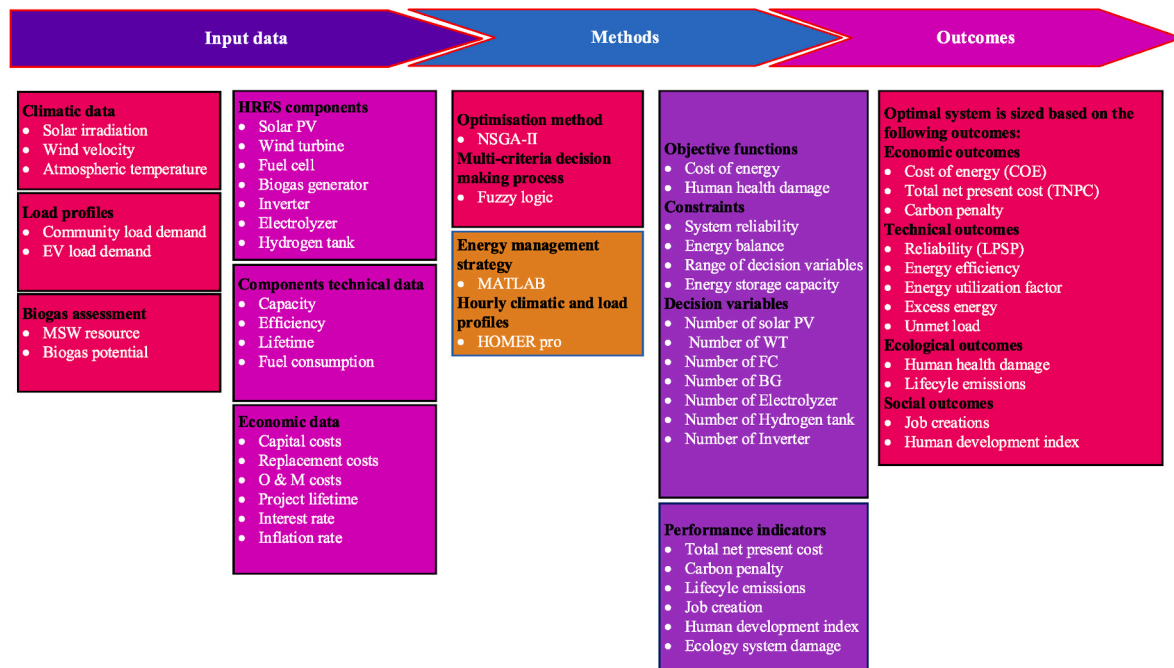


Fig. 1. Graphical representation of performance indicators considered in this study.

City Corporation (RCC), with a particular focus on the Katakhalī municipality. As depicted in the maps, RCC represents a significant urban area in northwestern Bangladesh. The leftmost map details the RCC's infrastructure, including key urban elements like roads, and municipal boundaries, which provides a good understanding the geographical context of the study area.

Katakhalī municipality (24°21'54.5"N 88°40'30.8"E), at the outskirts

of RCC, has been identified as the area with the significant concentration of grid-powered battery-operated electric vehicle (three-wheelers) charging stations. This concentration of charging stations makes Katakhalī an ideal candidate for the establishment of an off-grid power station, which has the potential to subsequently reduce the dependency on grid electricity.

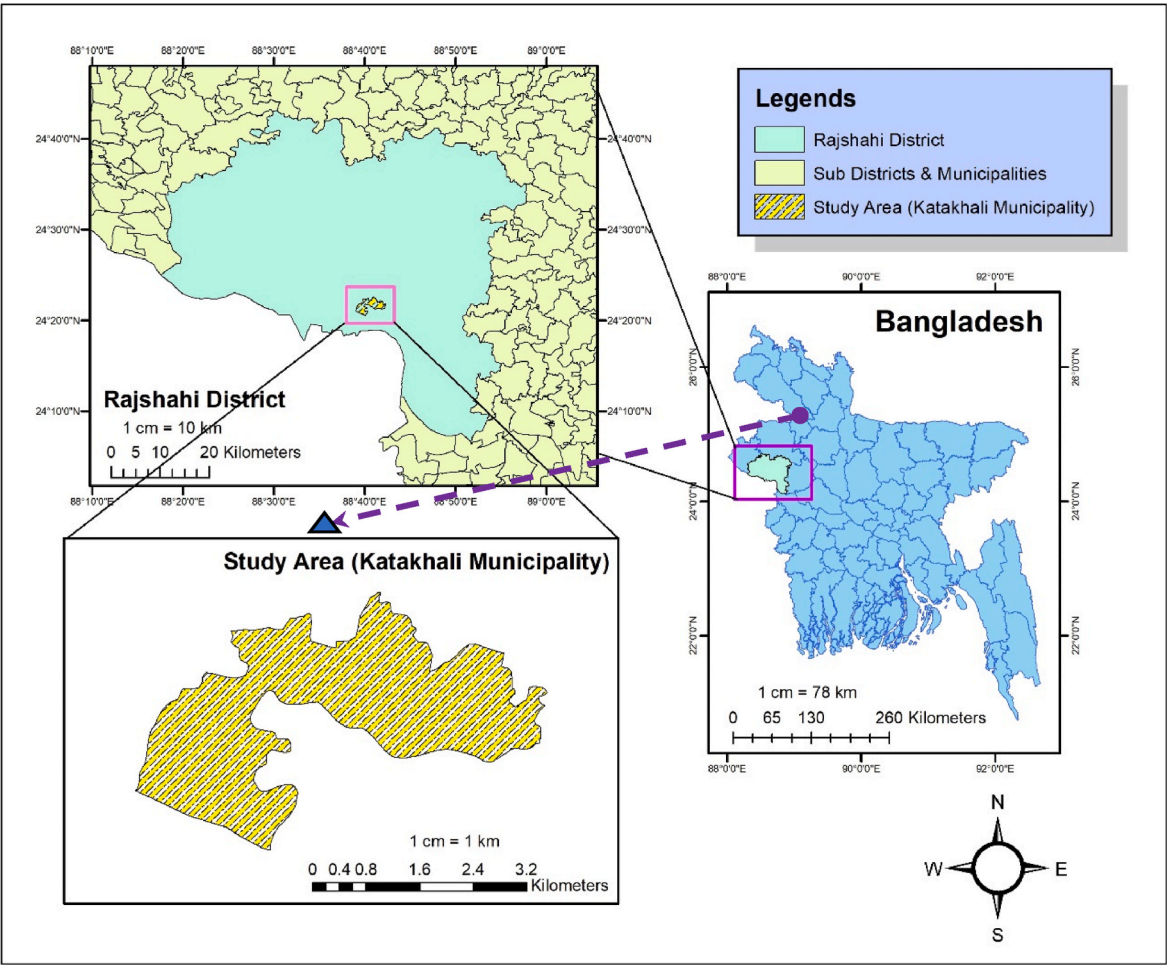


Fig. 2. Study area selected for this research.

2.1.1. Solar energy potential

The chosen study site boasts an ample supply of renewable energy resources. To acquire accurate data regarding solar irradiance and wind velocity for the chosen location, comprehensive statistics were sourced from the meteorological database of NASA within HOMER software tool. Fig. 3 presents a detailed time series of solar irradiation data for Katakhali, recorded in hourly time steps. The fluctuating nature of the solar irradiation, with peaks reaching close to 10 kW/m²/day and a daily average above 5 kW/m², showcases the potential for harnessing solar power effectively in this region.

2.1.2. Wind energy potential

Fig. 4 illustrates the corresponding wind speed data for the studied

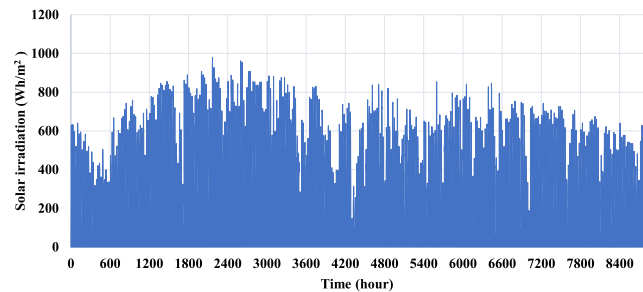


Fig. 3. Solar irradiation data for Katakhali in hourly time steps.

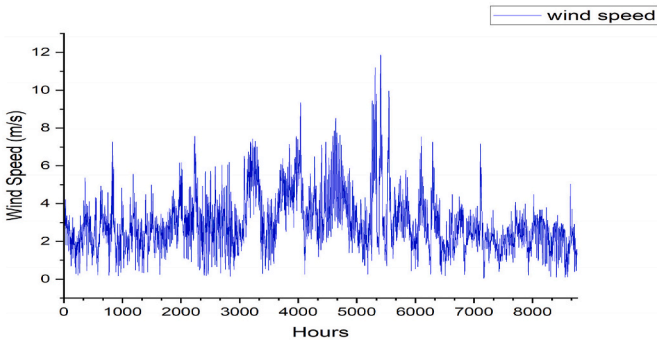


Fig. 4. Wind speed data for Katakhali in hourly time steps.

area. Notably, the average annual wind speed in the area stands at approximately 2.60 m/s which is the reason behind using vertical axis wind turbines for generating power output even at low wind speed.

2.1.3. Biogas energy potential

The study area considered for this research is a sub-urban municipality with enriched biogas feedstock resources which include biodegradable municipal solid waste (MSW), agricultural waste, and cattle manure mainly as cow dung. Katakhali generates plentiful amounts of cattle manure and biodegradable municipal solid waste enough to produce a reliable amount of biogas for power production. In this present study, MSW and cow dung are taken as feedstock sources for biogas

production. In Katakhal, it is estimated to be on average 1 cow per family, and around 15 kg of cow dung is produced on a daily basis per head of cow [16]. This shows that the study area can generate around 7500 kg/day of cow dung as raw feedstock for biogas production. It is reported that 1 kg of cow dung can produce 0.04 m³ of biogas which means to produce 1 m³ of biogas around 25 kg of cow dung is required as raw feedstock material [51], and around 300 m³ of biogas can be produced per day in the study area. The chemical composition of biogas includes 50–70% CH₄, 30–50% CO₂, and a small amount of H₂S as well and biogas has a lower heating value (LHV) in the range of 23 MJ/m³ with a density of 1.20 kg/m³ [52]. The cost of biogas is taken at 0.33 \$/m³ in the study [53]. The annual cattle dung production can be estimated from the following equation of M where K_j is the total number of cattle, m_j refers to per head dung production (ton), and k is the group of cattle (here cow only). The biogas generation (m³) from the used feedstock is determined by the equation of V_b [54], where S_{DMi} and S_{OMi} are the dry matter content in the dung (0.18) and the organic matter content (0.86) in the dry matter, respectively and V_{bi} is the specific biogas (211.86 m³/ton) from the organic matter [55].

$$M = \sum_{k=1}^j K_j m_j \quad (1)$$

$$V_b = \sum_{k=1}^j K_j \times m_j \times S_{DMi} \times S_{OMi} \times V_{bi} \quad (2)$$

Biodegradable MSW is another source of feedstock for biogas production. Aside from the selected off-grid community in Katakhal, Rajshahi City Corporation (RCC) is enriched with fruit & vegetable waste, food waste, and other biodegradable components which can be utilized to maximize generation. The following Table 2 indicates the different contents of MSW generation (Tons/day) in and around the study area and the area of Rajshahi City Corporation [56].

Using the following mathematical equation the present (2023) and future generation of MSW in RCC can be estimated which include different socio-economic and statistical factors ensuring enhanced estimation of MSW generation [57].

$$SWG_F = \frac{365}{1000} \times \left[G_c \left(1 + \frac{X_f}{100} \right)^{F-C} \right] \times \left[P_c \left(1 + \frac{Y_f}{100} \right)^{F-C} \right] \quad (3)$$

$$G_c = \frac{\text{Total landfill waste (in kg)}}{\text{Total population} \times \text{Time (Day)}} \quad (4)$$

The annual municipal solid waste generation can be estimated using equation (3) where the amount of waste generated is denoted by SWG_F in tons per year. MSW generated per capita per day is denoted by G_c and P_c is the total population in the corresponding municipal area. X_f and Y_f denote annual increased rate of per capita waste generation and annual increased rate of population in future respectively. C and F stand for present and future year respectively.

The following Fig. 5 provides the estimation of current and future generations of MSW among which around 71.1 % are counted as organic fractions for use as raw feedstock materials for the anaerobic digestion

Table 2

Different physical components of MSW on types of generation basis [56].

Waste type	Amount (tons/day)	Amount (%)
Foods & vegetables	88.31	71.1
Paper	11.05	8.9
Polythene & Plastic	4.97	4.0
Textile & Woods	2.36	1.9
Rubber & Leather	1.37	1.1
Metal & Tins	1.37	1.1
Glass & Ceramics	1.37	1.1
Brick, stone & Concrete	3.6	2.9
Others	1.62	1.3

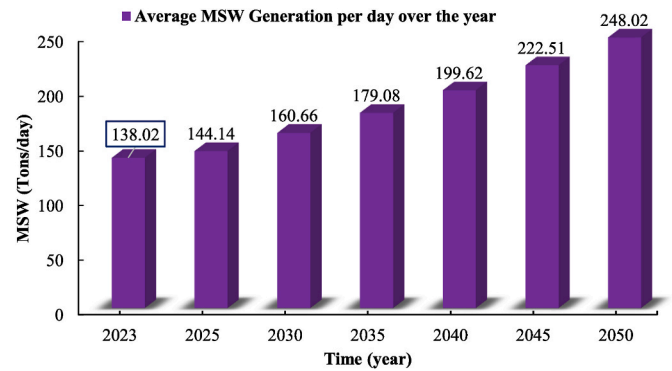


Fig. 5. Daily MSW generation over the years.

(AD) process to produce biogas.

In accordance with the investigation and practical assessment of diverse properties of Municipal Solid Waste (MSW), it has been established that 1 ton of MSW can generate 66 m³ of biogas [58]. Consequently, within the study area and Rajshahi City Corporation, an organic fraction of 98.13 tons of MSW is available, which can yield an impressive 6476.58 m³/day of biogas through the AD biogas plant. Therefore, including the generation potential of cattle manure, around 6776.58 m³ of biogas can be produced in total. Each cubic meter of biogas can be harnessed to produce approximately 6.39 kWh of electrical energy [59], culminating in a noteworthy total of 43302.35 kWh of power supply capability per day. This substantial biogas quantity is sufficiently potent to operate nearly 36 biogas generators, each having a rated power output of 50 kW. Notably, the actual number of generators utilized in an HRES could potentially exceed this figure, considering the intermittent nature of their usage.

2.2. Load assessment of EVCS and community supply

Table 3 presents the daily electrical energy consumption demands of the charging stations in the selected location. The aggregate load demand of Electric Vehicle Charging Stations (EVCS) remains relatively constant regardless of the season, with values of 3072 kWh per day and 2688 kWh per day observed throughout the summer and winter periods.

In Table 4, the community load estimation encompasses a total of 500 residences, two educational institutions, a network of streetlights, one main health center, ten small retail establishments, one post office, one bank, one commercial center, and two ATM booths. In both summer and winter timeline, the aggregate consumption load on a daily-basis amounts to 2461.06 kWh and 1837.34 kWh in the community, correspondingly.

2.3. Hybrid system architecture

This research introduces a novel HRES that incorporates solar PV modules, vertical axis wind turbines, fuel cells, electrolyzers, hydrogen-based energy storage units, and biogas generators. Managing variable renewables and accommodating fluctuating EVCS and community load demand simultaneously, this system ensures reliable and sustainable energy supply. Since the excess energy covers mainly the load demand of EVCS simultaneously with the community load, there is no chance of grid overload caused by EVCS, which happens in central power grid-based charging stations. The system architecture is shown in Fig. 6.

Solar photovoltaic panels and wind turbines collect energy, while hydrogen storage facility stores excess energy as hydrogen through electrolyzers for immediate use. A reserve biogas generator uses biogas generated from organic waste to generate electricity in a cost-effective manner. This comprehensive strategy assures a continuous energy supply by maximizing the use of renewable resources and incorporating a backup solution.

Table 3

Estimation of EVCS load requirements.

EVCS Load Demand						
Charging Stations	Summer (March–October)				Winter (November–February)	
	BEV Rating (kW)	Number of BEV charged per day	Charging duration (hour)	Daily Consumption (kWh/day)	Charging duration (hour)	Daily Consumption (kWh/day)
EVCS 01	4.8	13	8	499.20	7	436.80
EVCS 02	4.8	16	8	614.40	7	537.60
EVCS 03	4.8	20	8	768.00	7	672.00
EVCS 04	4.8	11	8	422.40	7	369.60
EVCS 05	4.8	20	8	768.00	7	672.00
Total load demand of EVCS (kWh/Day)				3072.00		2688.00

Table 4

Estimation of electric load requirement.

Community Load Demand						
Houses (500)						
Equipment	Summer (March–October)				Winter (November–February)	
	Rated Power (W)	Utilization Count	Operational hour (hr)	Daily Consumption (Wh/day)	Operational hour (hr)	Daily Consumption (kWh/day)
Fan	70	2	7	980	0	0
CFL bulb	25	4	7	700	8	800
TV	100	1	6	600	6	600
Refrigerator (33% duty cycle)	150	1	8	1200	8	1200
Mobile charger	6	2	3	36	3	36
Water Pump	746	1	1	746	1	746
Total (Wh/day) 1 house				4262		3382
Demand for 500 houses (kWh/day)				2131		1691
Educational institution (2) + primary health Care Centre (1)						
Equipment	Summer (March–October)				Winter (November–February)	
	Rated power (W)	Utilization count	Operational hour (hr)	Daily consumption (Wh/day)	Operational hour (hr)	Daily consumption (kWh/day)
Fan	70	110	7	53900	0	0
Air Conditioner	450	10	7	31500	0	0
CFL bulb	25	165	7	28875	8	33000
Computer	100	11	8	8800	8	8000
Water Pump	746	3	1	2238	1	2238
Refrigerator (33% duty cycle)	150	3	8	3600	8	3600
Total (Wh/day) 2 School & college				128913		46838
Total demand (kWh/day)				128.91		46.84
Commercial center (1) + small retail establishments (10) + post office (1) + bank (1) + ATM booth (2)						
Equipment	Summer (March–October)				Winter (November–February)	
	Rated power (W)	Utilization count	Operational hour (hr)	Daily consumption (Wh/day)	Operational hour (hr)	Daily consumption (kWh/day)
Fan	70	110	7	53900	0	0
Air Conditioner	450	20	6	54000	0	0
CFL bulb	25	250	7	43750	8	50000
Computer	75	50	6	22500	6	22500
Refrigerator (33% duty cycle)	150	15	8	18000	8	18000
TV	100	15	6	9000	6	9000
Total				201150		99500
Total demand (kWh/day)				201.15		99.50
Grand total load demand of community (kWh/day)				2461.06		1837.34

2.4. Mathematical modelling

Photovoltaic: The photovoltaic module's output power can be calculated by using the following formula [60].

$$W_{PV}(t) = W_{RPV} \times f_{PV} \times \left(\frac{I_T}{I_s} \right) [1 + \mu(T_C - T_s)] \quad (5)$$

where W_{RPV} denotes the maximum power output of the PV cell, f_{PV} is defined as the degradation coefficient, I_T defines the solar irradiance at specific temperature of the environment, I_s signifies the solar irradiation measured under standardized test condition, the coefficient of temperature is denoted by μ . Lastly, T_C and T_s represent the temperature of PV cell in that environment and at standard test conditions, respectively.

The estimated cell temperature can be found using the following

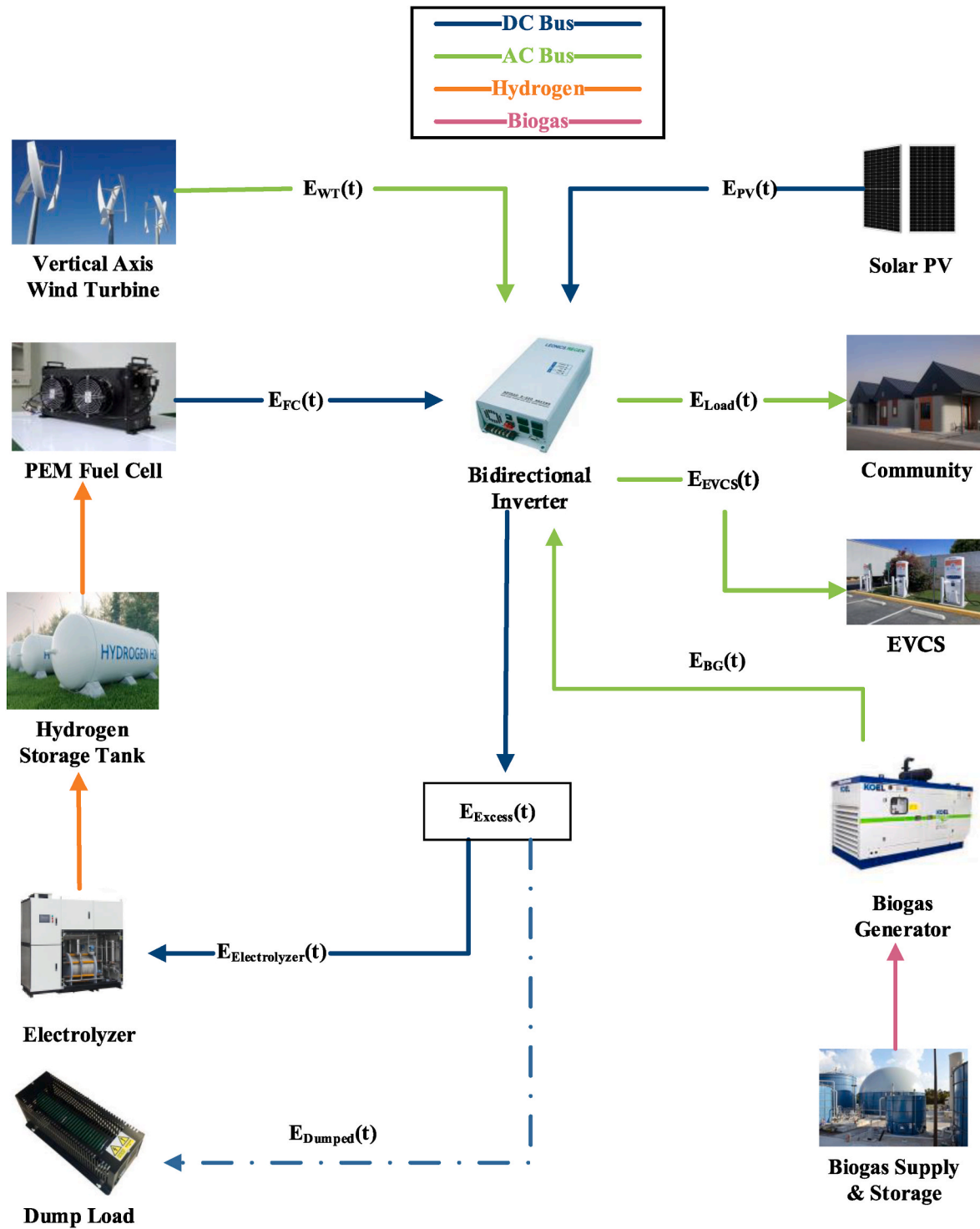


Fig. 6. Designed system architecture of HRES.

equation below, $T_{C,NOCT}$, $T_{a,NOCT}$, and $I_{T,NOCT}$ are used for nominal operating cell temperature (NOCT), covering ambient temperature at NOCT and solar irradiation at NOCT, respectively. η_{PV} represents PV panel efficiency which is considered to be 21.4% according to the selected type of the cell [61].

$$T_C = T_a + I_T \frac{(T_{C,NOCT} - T_{a,NOCT})}{I_{T,NOCT}} \times \left(1 - \frac{\eta_{PV}}{0.9}\right) \quad (6)$$

Wind turbine: The low wind energy potential of the study area selected for this research has made it conducive to consider H-type

Darrius vertical axis wind turbine (VAWT) for this research. It is highly efficient for low wind speed power production which provides more effective swept area than horizontal axis wind turbine (HAWT). The generator efficiency of VAWT considered is 80% and the turbine efficiency of the VAWT is comparatively higher than HAWT in low wind velocity. The power output of a wind turbine can be obtained from the following equation [62].

$$P_{WT} = \begin{cases} 0; & V \leq V_{cut\ in} \\ a \times V^3 - b \times W_{rated}; & V_{cut\ in} \leq V \leq V_{rated} \\ W_{rated}; & V_{rated} \leq V \leq V_{cut\ off} \\ 0; & V > V_{cut\ off} \end{cases} \quad (7)$$

$$a = \frac{W_{rated}}{V_{rated}^3 - V_{cut\ in}^3} \quad (8)$$

$$b = \frac{V_{cut\ in}^3}{V_{rated}^3 - V_{cut\ in}^3} \quad (9)$$

Here, P_{WT} is the power output, W_{rated} is the rated power in kW, $V_{cut\ in}$, V_{rated} and $V_{cut\ off}$ are the starting, rated, and cut-off velocities (m/s) of the VAWT. Determination of the actual power output from the wind turbine is expressed as: $P_{WT,electrical} = P_{WT} \times S_{swept} \times \eta_{WT}$.

Where the total swept area for a VAWT is $S_{swept} = D \times H$. D is the wheel diameter, H is the rotor height of VAWT and η_{WT} is the efficiency of wind turbines [63].

Biogas generator: The electrical energy (kWh) from Biogas Generator can be estimated using equation (10). This equation is related to the conversion of biogas to useful energy in the form of electricity and heat [64].

$$E_{BG} = \frac{LVH \times Q_{br} \times \eta}{\gamma_i} \quad (10)$$

E_{BG} , LVH , Q_{br} , η , γ_i denote the available electrical energy (kWh), the lower biogas calorific value, the recoverable biogas flow (m^3), the electrical efficiency of the generating element (Biogas Generator) in transforming thermal energy into electrical energy and the conversion factor of MJ into kWh (1 MJ/0.28 kWh), respectively. The overall efficiency of the biogas generator considered in this study is 28.45%. The power of the generator can be determined from the following equation.

$$P = \frac{E_{BG}}{8760 \times CF} \quad (11)$$

Where P is the power of the generator (kW) and CF , or capacity factor, is the plant availability factor, which is commonly assumed to be between 80% and 90%.

Fuel cell: The energy supplied by the fuel cell (FC) at a certain time-step is determined as follows:

$$E_{FC}(t) = \frac{[(E_{Elec}(t) + E_{EVCS}(t)) - (E_{PV}(t) + E_{WT}(t))]}{\eta_{inv}} \quad (12)$$

The amount of hydrogen needed to be converted by the fuel cell, H_{FC} (kg), to resolve the energy shortfall, is calculated using the equation below at any time step t .

$$H_{FC}(t) = \frac{E_{FC}(t)}{\eta_{FC} \times HHV_{H_2}} \quad (13)$$

η_{FC} and HHV_{H_2} are the electrical conversion efficiency of FC (50%) and the calorific value of hydrogen, respectively (39.4 kWh/kg) [65].

Electrolyzer: The amount of extra electrical energy that is supplied to the electrolyzer and transformed into hydrogen at each time step can be calculated as follows [65]:

$$E_{Electrolyzer}(t) = [E_{PV}(t) + E_{WT}(t) - (E_{Elec}(t))] \times \eta_{inv} \quad (14)$$

In an electrolyzer, electrical energy is transformed into hydrogen, and the amount of hydrogen generated each hour $H_{Electrolyzer}$ (kg), can be calculated as follows [65]:

$$H_{Electrolyzer}(t) = 1.43 \times 10^{-3} + 2.39 \times 10^{-2} E_{Electrolyzer}(t) - 4.32 \times 10^{-5} E_{Electrolyzer}^2 \quad (15)$$

Hydrogen storage: The storage module of hydrogen holds any

additional energy produced by the renewables as H_2 . This H_2 is produced by running electrolyzer with the help of excess energy from the renewables. At any time-step, when the total renewable production surpasses the load demand, the following equation becomes the one of interest:

$$H_{HS}(t) = H_{HS}(t-1) + H_{Electrolyzer}(t) \quad (16)$$

$H_{HS}(t)$ and $H_{HS}(t-1)$ represent the total amount of hydrogen (kg) kept in storage at time steps t and $(t-1)$. Via a fuel cell, the hydrogen that has been stored is transformed into electrical energy. Under these circumstances, the amount of hydrogen kept in storage tank is calculated as follows:

$$H_{HS}(t) = H_{HS}(t-1) + H_{FC}(t) \quad (17)$$

$H_{FC}(t)$ denotes the hydrogen quantity withdrawn from storage and converted into electrical power to meet the deficit of load demand.

Bi-directional inverter: With the bidirectional inverter, power can be converted bidirectionally, switching between DC and AC whenever required. The output power $P_{out(t)}$ at the load side after conversion can be determined by the following equation [17].

$$P_{out(t)} = \eta_{inv} \times P_{in(t)} \quad (18)$$

Where, $P_{in(t)}$ denotes the input power at any time and η_{inv} signifies the efficiency of the inverter.

2.5. Power management strategy

Fig. 7 shows the operational control algorithm of the hybrid renewable energy system (HRES). The control algorithm determines the demand-generation status based on input factors like electric load requirements, renewable energy availability, and temperature data. It also controls the energy storage techniques described below:

- **Renewable energy generation exceeds the energy demand, $E_{Net}(t) > 0$:** In this scenario, the demand for the electric load has been completely met, and there is excess energy ($E_{Excess}(t) = E_{Net}(t) - E_{Elec}(t)$). The algorithm checks another condition ($E_{Excess}(t) > E_{EVCS}(t)$). If the condition is true, the algorithm advances as the extra energy is put to work generating hydrogen through electrolysis. The H_2 tank holds the produced hydrogen to the fullest extent possible. When the H_2 storage tank is full, the extra energy equivalent to that amount of H_2 is discharged through a resistive load. If $E_{Excess}(t) < E_{EVCS}(t)$, there is a visible deficit to meet the EVCS demand. The way to make up for that deficit takes us to the next section.
- **The generation of renewable energy balances the demand, $E_{Net}(t) = 0$:** In this instance, the renewable energy production (solar, wind) meets the electrical load demand and there is no excess energy produced. To meet the EVCS load demand or the deficit, the H_2 tank's SOC is checked. If it has sufficient hydrogen to run the fuel cell, the EVCS demand will be met. Otherwise, the biogas generator will act as the energy supplier. However, the threshold power of the biogas generator considered here is 30% of the maximum power as the minimum power of a single unit is taken as 15 kW. The demand of energy consumption below this percentage threshold will be considered an "unmet load".
- **Renewable energy generation falls short of the demand ($E_{Net}(t) < 0$):** If there is any renewable energy available in this scenario, after using it, the fuel cell makes up the difference in the load requirement, and the hydrogen is used from the H_2 Tank. Given that the load demand is equal to or more than 30% of the rated power, the biogas generator fills the energy gap if the deficiency persists. Any load demand that is less than this mark is regarded as an unmet load. When employing renewable energy sources and energy storage is not

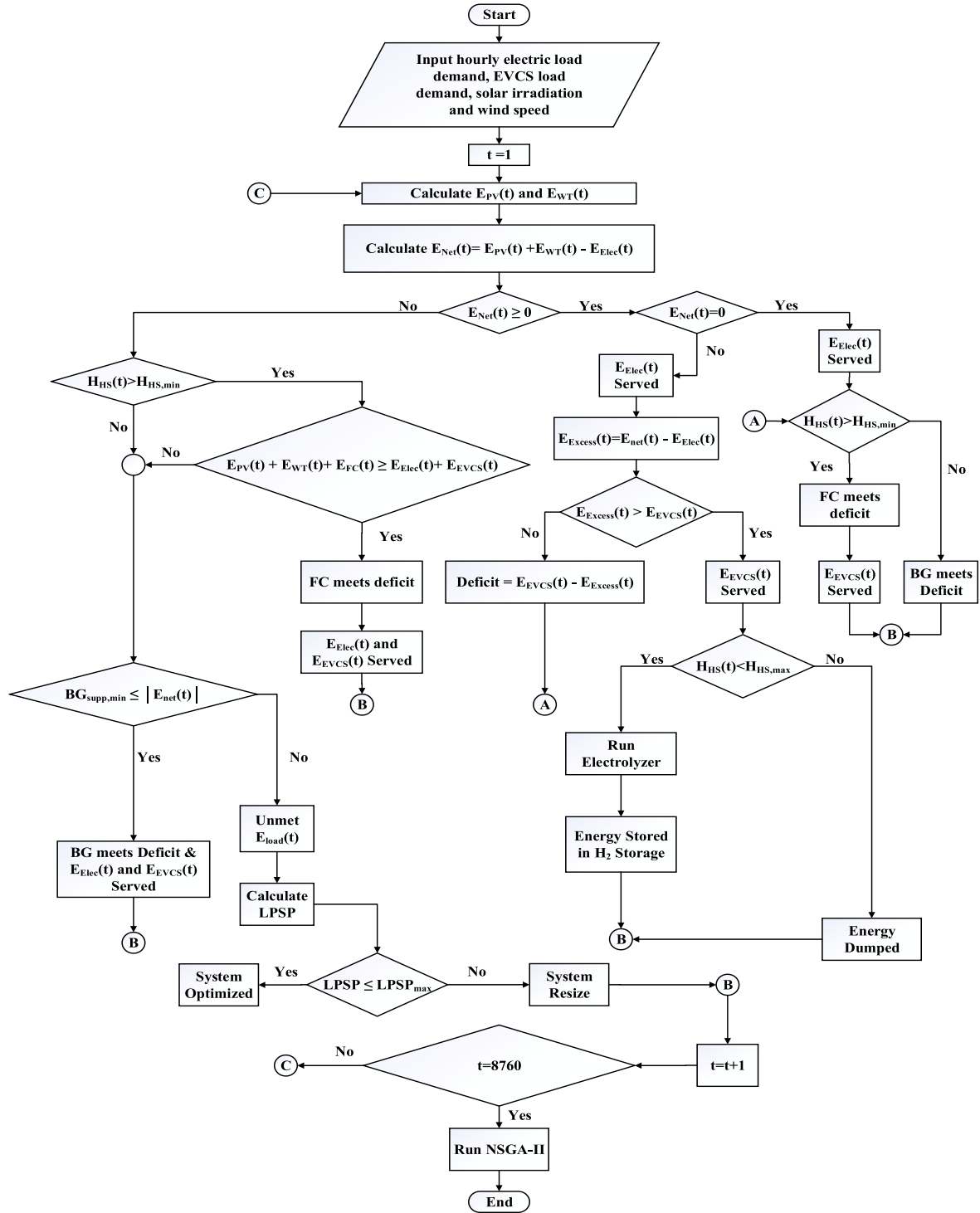


Fig. 7. Power management strategy for hybrid system for meeting electrical and EVCS load demand.

an option, the biogas generator runs at maximum capacity to satisfy the demand.

2.6. Performance parameters of hybrid system

Net present cost (NPC): The net present cost can be estimated as follows [66]:

$$NPC = C_{cap} + C_{O\&M} + C_{rep} - C_{salvage} \quad (19)$$

Moreover, the following equations can help us determine the capital

cost (C_{cap}), operation and maintenance cost ($C_{O\&M}$), replacement cost (C_{rep}), and salvage value ($C_{salvage}$) for each component;

$$Capital\ Cost : C_{cap} = N_{comp} \times C_{cap,comp} \quad (20)$$

$$Replacement\ Cost\ C_{rep} = N_{comp} \times C_{rep,comp} \sum_{j=1}^{N_{rep-comp}} \frac{1}{(1+i)^{L_{comp} \cdot j}} \quad (21)$$

$$\text{Operational and Maintenance Cost : } C_{O\&M} = N_{comp} \times C_{o\&m,comp} \sum_{k=1}^L \frac{1}{(1+i)^k} \quad (22)$$

$$\text{Salvage Value : } C_{sal} = C_{rep,comp} \times \frac{R_{rem,comp}}{L_{comp}} \left(\frac{1}{(1+i)^{L_{comp}}} \right) \quad (23)$$

Where $C_{cap, comp}$ denotes the cost of initial purchase and $C_{rep, comp}$ denotes the cost of final replacement, N_{comp} denotes the total number of components in the ideal hybrid system.

The $N_{rep-comp}$ metric calculates the frequency of component replacement. $R_{rem, comp}$ stands for a component's remaining useful life. L_{comp} is the lifetime of system component, $C_{O\&M, comp}$ is the lifetime index of the project, and k is the component's cost of operation and maintenance.

Cost of energy (COE): Calculating the cost of energy (COE) for a hybridized project requires the following formula [67]:

$$\text{Min COE} = \frac{NPC \times CRF(i, L)}{\sum_{t=1}^{8760} E_s} \quad (24)$$

Where E_s is the yearly load supplied, $CRF(i, L)$ is the capital recovery factor, and NPC represents the net present cost of the hybrid system components.

$CRF(i, L)$ is calculated by using the below equation [68]:

$$CRF(i, L) = \frac{i(1+i)^L}{(1+i)^L - 1} \quad (25)$$

where i is the real annual interest rate and L is the lifetime of the project.

Lifecycle emission: The following Equation can be used to calculate the life cycle emissions for a given year of the HRES [69].

$$LCE = \sum_m \sum_{i=1}^T \psi_m E_{m,i}, m \in \{PV, WT, BAT, BG, FC, Electrolyzer\} \quad (26)$$

Where ψ indicates the cumulative CO_2 emissions over its lifespan, and E represents the stored or converted energy (via FC, electrolyzer, battery, PV modules, wind turbines, biogas generators, and inverters) needed to meet the load demand throughout T (8760 h).

Human health and ecosystem damage: The system's components will emit carbon dioxide into the atmosphere, which will contribute to an increase in the average global temperature, affecting ecosystems and the well-being of humans [70]. Disability-adjusted life years (DALYs) measure the number of years lost owing to disability or early mortality to quantify the harm done to human health (HHD). In this study, the human health damage factor has been taken as $1.4E-06$ DALY/kg CO_2 -eq. Ecosystem damage (ESD), which is the total number of species destroyed in a given area over a specific period, is measured in species years and the factor taken is $7.93E-09$ species.yr/kg CO_2 -eq [71].

$$HHD = LCE \times 1.40 \times 10^{-06} \quad (27)$$

$$ESD = LCE \times 7.93 \times 10^{-09} \quad (28)$$

Human development index: The human development index, or HDI for short, evaluates the general level of socioeconomic development of people. Apart from that, it can be noticed in the tendencies of consumption of energy that people in a certain area typically display. In this scenario, local small companies can likely make use of the yearly excess energy, potentially enhancing both the HDI and the quality of life. Following relation can be used to determine the HDI [72].

$$HDI = 0.0978 \ln \left[\frac{E_{load} + \min(f_{ex,max} \times E_{ex}, f_{load,max} \times E_{load})}{\text{Number of People}} \right] - 0.0319 \quad (29)$$

Where E_{load} and E_{ex} are shorthand for the system's AC load on an annual basis and energy surplus over the year respectively. The values of the factors $f_{ex, max}$, and $f_{load, max}$ are considered to be 0.2 and 0.5, respectively [73].

Job creation (JC): The probability of created job opportunities is demonstrated by the job creation (JC) factor, which helps to contribute to the reduction of unemployment in a particular place. The following equation can be used to make an approximation of the job creation (JC) factor [72]:

$$JC = JC_{PV} \times P_{PV} + JC_{WT} \times P_{WT} + JC_{BG} \times P_{BG} + JC_{FC} \times P_{FC} \quad (30)$$

In this particular scenario, JC_{PV} , JC_{WT} , JC_{BG} , and JC_{FC} are presented as the respective job quantities per MW for PV energy, wind turbines, biogas generators, and fuel cells, respectively. These are the main power generation sources of the HRES. The associated values show how many jobs are related to each energy source or plant. Table 5 serves as a thorough reference for the relevant data and provides access to key input parameters that will help with the estimation of the JC factor.

Excess energy utilization factor: The energy utilization factor (EUF) is a crucial metric for evaluating efficiency in a HRES, focusing on the optimal use of surplus energy and minimizing waste. This factor is determined by calculating the ratio of the difference between excess and dumped energy to the excess energy itself. High EUF in HRES enhances micro-grid supply stability and guides the appropriate sizing of energy storage systems. Essentially, the EUF serves as an efficiency gauge for the HRES, ensuring maximum utilization of renewable energy resources and optimizing energy storage and can be calculated using the following relation

$$\text{Energy Utilization Factor, EUF} = \frac{\text{Excess Energy} - \text{Dumped Energy}}{\text{Excess Energy}} \quad (31)$$

Incremental load condition and system stability criteria: In the HRES, it is essential to account for incremental load situations, which involve gradual increases in energy demand caused by factors such as population growth and technological advancements. This consideration is crucial for ensuring the system's stability and its long-term sustainability. Neglecting these changes can lead to inadequate capacity planning, risking system overload, blackouts, or reliance on non-renewable backups, thereby compromising sustainability goals. Furthermore, incremental loads significantly impact system efficiency and the excess energy fraction (EEF), key metrics for assessing HRES performance [2]. The following equations are of particular interest.

$$\text{HRES Efficiency (\%)} = \frac{\text{Total Generation} - \text{Dumped Energy}}{\text{Total Generation}} \times 100 \quad (32)$$

3. Optimization framework

In this study, the multi-objective functions and the constraints are developed in MATLAB environment based on the energy balance and PMS described in Fig. 7. The climatic information and load data are developed in HOMER software for the hourly temporal resolution and then used in the MATLAB M-files. The multi-objective optimization and the multi-criteria decision-making process are performed in the MATLAB environment.

Table 5
JC factor estimation parameters [74–76].

System Components	JC Factors
PV module	2.7 jobs/MW
Wind turbine	1.1 jobs/MW
Fuel cell	1.2 jobs/MW
Biogas generator	1.64 jobs/MW

3.1. Objective functions

During the development of an optimization process, the research considered multiple factors. These factors include minimizing the cost of energy (COE) and the adverse effects on human health (HHD) while ensuring the delivery of electricity within a specified level of load reliability (LPSP = 1 ± 0.5). Consequently, the objective function can be defined as follows:

$$F = \text{minimize } \{COE, HHD\} \quad (33)$$

The COE functions as a widely acknowledged economic metric that investors frequently use to ascertain the appropriate sizing of hybrid energy system. It offers increased transparency, making it a valuable tool [77]. However, assessing the lifecycle CO₂ emissions serves as a crucial indicator for evaluating the environmental sustainability of deploying hybrid energy systems, contributing to overall sustainable development.

Focusing solely on quantifying GHG emissions fails to adequately capture the impact on everyday individuals, especially those residing in rural and suburban areas. This research aims to minimize the damage to human health as an additional objective function when optimizing the suggested hybrid renewable systems. Such approach seeks to address public concerns about their well-being, promoting the acceptance of these systems. Moreover, it strives to improve societal awareness regarding the environmental and health outcomes tied to the present utility methods in the study area.

3.2. Constraints

Range of decision variables: When optimizing the size of a HRES, the number of system components becomes a crucial criterion to consider. To ensure efficient calculations during the optimization process, predetermined upper and lower ranges are imposed as constraints. These constraints are defined as follows:

$$N_{m,min} \leq N_m \leq N_{m,max}, m \in (PV, WT, BG, FC, Inv, H_2 \text{ Tank}, Electrolyzer) \quad (34)$$

Where N_m is the total number of m -th system components, $N_{m,min}$ is the smallest possible number of m th system components and $N_{m,max}$ is the largest possible number of m -th system components. To speed up computation, the solution space is constrained by the upper and lower bounds. The minimum number of components are taken as 1, while the upper bounds are as follows:

$$N_{pv} = 2000, N_{WT} = 180, N_{FC} = 300, N_{H_2 \text{ tank}} = 400, N_{electrolyzer} = 300, N_{BG} = 9$$

These values have been found after implementing the NSGA-II and fuzzy algorithms on MATLAB. After carrying out the procedure on a basis of trial and error, this is figured out during an initial stage of the optimization process.

Constraints of energy balance: The cumulative energy output from all system components per hour must meet or exceed the hourly load demand.

$$E_{pv}(t) + E_{WT}(t) + E_{BG}(t) + E_{FC}(t) - E_{Electrolyzer}(t) = E_L(t) + E_{EVCS}(t) + E_{Excess}(t) \quad (35)$$

E_{pv} and E_{WT} are the renewable or primary sources of power to meet load demand while E_{BG} , E_{FC} , and $E_{Electrolyzer}$, are the energy delivered as backup and stored while fulfilling the overall energy demand $E_L(t)$ at a certain hour. E_{pv} , E_{WT} , E_{FC} , and E_{BG} are the energy produced from PV, WT, FC, and BG, respectively.

3.2.1. Energy storage capacity constraints

The following limitation governs the amount of hydrogen that can be

stored in the system:

$$H_{HS,min} < H_{HS}(t) < H_{HS,max} \quad (36)$$

Here the terms $H_{HS,min}$ and $H_{HS,max}$ represent the minimum and maximum capacities (in kilograms) of the hydrogen energy storage system.

Constraints of system reliability: The reliability indicator is one of the most crucial factors while designing HRES. The loss of power supply probability (LPSP), commonly used as a reliability indicator, is used to evaluate the dependability of the HRES. This index measures the likelihood that the system would fail to meet the over the course of a year (8760 time-steps or hours), load demand. The following equation is used to calculate the LPSP [60]:

$$LPSP = \frac{\sum_{t=1}^T LPSP(t)}{\sum_{t=1}^T E_L(t)} \quad (37)$$

When backup, supplemental, and storage systems are unable to produce or provide enough power to fulfill demand, the LPSP is computed to account for the least favorable power supply condition. The equation may be used to determine it during T (8760 h):

$$\sum_{t=1}^T LPSP(t) = \sum_{t=1}^T (E_L(t) - [E_{Gen}(t) + E_{FC,max}(t)] \times \eta_{inv}) \quad (38)$$

The constraint on system reliability is then written as:

$$LPSP \leq LPSP_{Desired} \quad (39)$$

Where $LPSP_{Desired}$ denotes the system's necessary reliability and is taken to be $1 \pm 0.5\%$.

3.3. Optimization algorithm (NSGA-II)

The upgraded NSGA-II algorithm, developed for hybrid energy system optimization, addresses the limitations of the original NSGA. It features elitism and eliminates the need for a diversity-sharing parameter by using a crowding distance operator, leading to enhanced accuracy and faster computation. The overall complexity of NSGA-II is efficiently expressed as $O(MN^2)$, where M is the number of objectives and N the population size, marking a significant improvement over its predecessor [78].

While maintaining a diverse spread along the Pareto-optimal front, NSGA-II achieves effective convergence. In contrast to PSO, it excels at finding global optima and handling a wide range of parameters [79]. Deb et al. [80] provided a thorough understanding of the algorithm by carrying out the implementation of a sensitivity analysis to determine various optimization parameters. The following Table 6 presents the optimization parameters of NSGA-II.

3.4. Determination of final solutions NSGA-II

The concept of crowding distance is employed to gauge the concentration of solutions surrounding a chosen one, contingent on whether the objectives are being maximized or minimized. Essentially, it quantifies the average spacing between adjacent solutions along each

Table 6
Control variables of NSGA-II algorithm.

Parameters	Value
Size of population	200
Maximum generation	200
Rate of crossover (SBX crossover)	0.9
Rate of mutation (Poly mutation)	0.1

dimension of the objective space.

In this study, the NSGA-II algorithm has identified a range of non-dominant solutions on the Pareto front from a multi-objective optimization. Selecting the most favorable solution requires balancing trade-offs. Hence, a fuzzy decision-making approach is employed to determine the optimal choice from the Pareto front. The calculation of the fuzzy membership value for the j th objective function is performed as follows [81], where $F_j^{\max}$ represents the maximum fitness value of the j th objective function, and $F_j^{\min}$ denotes the minimum fitness value.

$$\chi_j = \begin{cases} 1 & \text{for } F_j \leq F_j^{\min} \\ \frac{F_j^{\max} - F_j}{F_j^{\max} - F_j^{\min}} & \text{for } F_j^{\min} < F_j < F_j^{\max} \\ 0 & \text{for } F_j \geq F_j^{\max} \end{cases} \quad (40)$$

For each non-dominant solution, the normalized membership function is defined as χ^k [81]:

$$\chi^k = \frac{\sum_{i=1}^{N_{obj}} \chi_j^k}{\sum_{k=1}^M \sum_{j=1}^{N_{obj}} \chi_j^k} \quad (41)$$

The normalized membership functions yield the final solution based on the highest χ^k value, where M stands for the non-dominated solutions and N_{obj} denotes the objective function number.

3.5. Assumptions

The optimization process of the proposed HRES includes the following assumptions:

- The rural area corresponding to the analysis of the HRES is considered as off-grid to avoid dependency on the grid and to ensure that the system can supply power to the considered community load and EVCS load simultaneously without any interference.
- Load profiles and climatic data are examined with hourly temporal resolution, under the assumption that variations within the hour are negligible.
- This study excludes consideration of power generation losses, HRES device failures, and transmission and distribution losses.
- Effects of humidity, ambient pressure, and soiling may affect the power out of a PV module to some extent. However, they were neglected to avoid convoluted calculations.
- The analysis of costs excludes considerations for civil structure, instrumentation, piping, and fabrication.

4. Results and discussion

This study examines the optimization of hybrid renewable systems consisting of solar PV, wind turbines, biogas generator, fuel cells, electrolyzer, and hydrogen storage for the concurrent provision of electricity to both the charging stations and the community. Three different HRES configurations are considered in this study and their summarized results are reported in Table 7. The optimized capacity of PV, WT, FC, and BG are 270 kW, 375 kW, 70 kW, and 500 kW, respectively for the PV/WT/FC/BG-based HRES system as reported in Table 7. Larger PV and FC capacities are required to meet the demand if the WT is excluded (PV/FC/BG). Similarly, higher capacity of WT is needed for the WT/FC/BG-based HRES to satisfy the load requirements (Table 7). Further breakdown can be found in the following sections.

The application of the multi-objective genetic algorithm NSGA-II results in distinctly shown Pareto fronts of the configurations, as

Table 7

Breakdown of component capacities and energy generation of optimized models.

System Components	PV/WT/FC/BG	PV/FC/BG	WT/FC/BG
PV module (kW)	270	842	0
Wind turbine (kW)	375	0	890
Biogas generator (kW)	500	500	500
FC (kW)	70	200	58
H ₂ tank (kg)	726	1246	1104
Electrolyzer (kW)	200	274	74
Converter (kW)	311	969	67
PV energy (kWh/yr.)	453,956	1,415,415	0
Wind energy (kWh/yr.)	235,895	0	558,388
Biogas energy (kWh/yr.)	1,400,252	1,083,146	1,581,642
Excess energy (kWh/yr.)	273,130	808,788	310,855
Total electrical demand (kWh/yr.)	1,907,186	1,907,186	1,907,186
Unmet electric demand (kWh)	16,692	19,838	10,128

shown in Fig. 8(a), (c), and (d). The final configuration size and the optimal solution (point shown in red) are then selected using a fuzzy-logic decision making algorithm. On the other hand, Fig. 8(b) demonstrates the convergence behavior of the NSGA-II optimization algorithm, specifically when the objective is focused solely on minimizing the COE. Fig. 8(a) illustrates the COE and HHD for the PV/WT/FC/BG-based HRES are 0.1634 \$/kWh and 0.1732, respectively. However, slightly lower COE can be achieved when the HRES is optimized for a single objective function of COE as reported in Fig. 8(b). The COE and HHD are comparatively higher for the PV/FC/BG and WT/FC/BG-based HRES as shown in Fig. 8(c) and (d), respectively.

4.1. Assessment of technical performance

4.1.1. Energy balance and system scaling of optimized hybrid systems

The power interaction scenarios of the three hybrid systems for a specific day (6 a.m. to next day 5AM), as shown in Fig. 9(a), (b), and (c), reveals how each component of the system operates to meet the combined load demand of the community and the charging stations. In the PV/WT/FC/BG system (Fig. 9(a)), the daytime is marked by high solar and wind potential, which covers the combined load demand. The excess energy is then used to run the electrolyzer to store hydrogen, as in this instance, the hydrogen tanks are far from reaching their maximum capacity (14254 kW). There is no dumped energy or unmet load in this particular span of time. At night and early morning, BG are crucial in meeting the high energy demand when renewable production is low or absent. Peak load occurs at 7 a.m. and it is met with the help of fuel cells as the biogas generators reach their maximum capacity, with lower wind contribution.

Without the wind turbines, PVs take a larger role, causing their size to increase significantly (Fig. 9(b)). As a result, there is an increase in the power generation from the PVs in the PV/FC/BG configuration. This leads to more surplus energy, some of which is used to run the electrolyzer and store hydrogen while a significant portion of energy is dumped. Although not visibly apparent, there is an instance of unmet load at 7 a.m. The biogas generator meets the electrical load demand, but the PV generation is insufficient compensate for the EVCS demand.

The exclusion of PV from the system leads to an overall increase in the size (134%) and the power output of the wind turbines (Fig. 9(c)) compared to the original. Surplus energy stays almost the same, but some instances of dumped energy are seen due to a lower share of electrolyzers. Compared to the PV/FC/BG system, the excess energy utilization rate is higher in the WT/FC/BG system, leading to a better system efficiency.

4.1.2. Efficiency analysis of optimized HRES

The efficiency of a HRES is defined by the ratio of energy supplied to total energy produced. Table 8 shows that PV/WT/FC/BG has the highest system efficiency with minimal loss. Energy losses mainly occur during AC to DC conversion and in the FC energy conversion process.

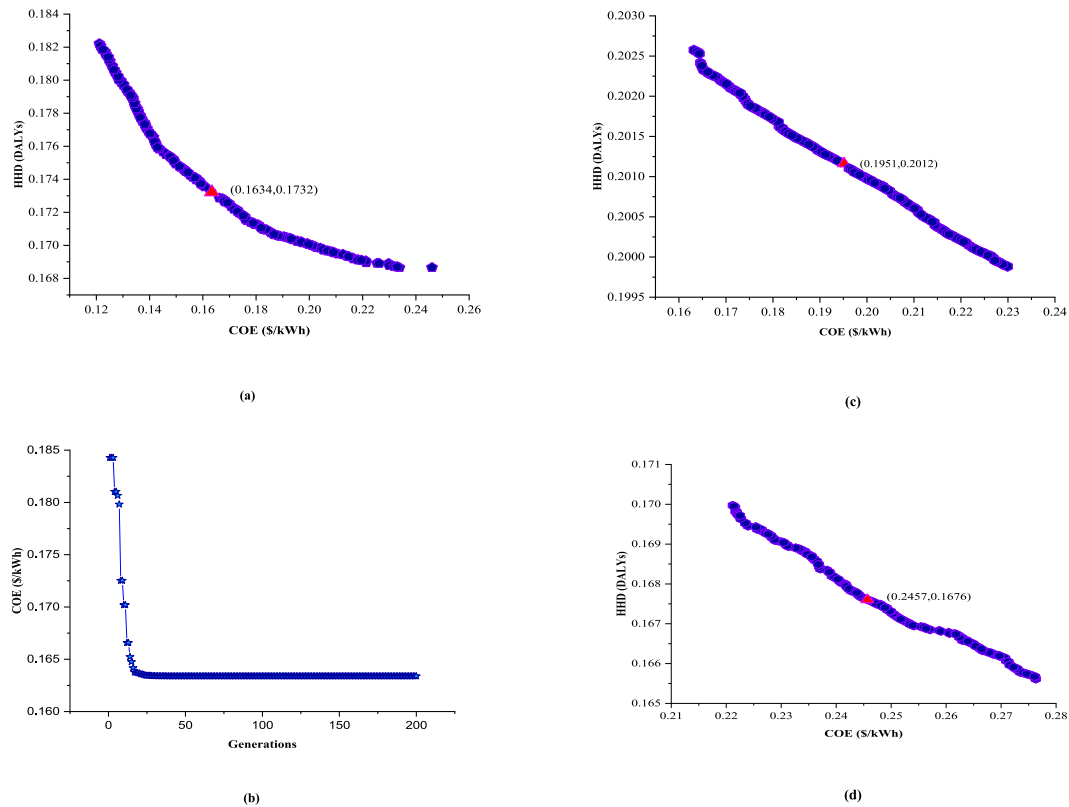


Fig. 8. Optimized Pareto front (a) for double objectives and (b) single objective of PV/WT/FC/BG system, (c) for PV/FC/BG system, and (d) WT/FC/BG system.

The WT/FC/BG system faces much higher losses than PV/WT/FC/BG and PV/FC/BG, largely due to the greater power output from its WTs and a lower share of electrolyzers. On the other hand, the PV/FC/BG system generates significantly more surplus energy, at 808,788 kWh annually, compared to 273,130 kWh for PV/WT/FC/BG and 310,855 kWh for WT/FC/BG. This difference is because the PV/FC/BG system relies more on PV energy. The combination of PVs and wind turbines in the PV/WT/FC/BG system leads to a more balanced generation profile, reducing the excess energy by better matching the energy supply with demand.

4.2. Cost analysis

The PV/WT/FC/BG-based HRES configuration offers the lowest COE of 0.1634 \$/kWh as seen in Fig. 10. The COEs for the PV/FC/BG and WT/FC/BG are 0.1951 \$/kWh and 0.2457 \$/kWh, respectively. The NPC for PV/WT/FC/BG-based option is \$3988169, whereas the NPCs for the PV/FC/BG and WT/FC/BG options are \$4762796 and \$5997060, respectively. Fig. 11 shows how each part of the three hybrid systems contributes to the NPC. Biogas generators share the largest share of total cost in PV/WT/FC/BG and PV/FC/BG systems, contributing up to 36.20% for PV/WT/FC/BG and 27.92% for PV/FC/BG. In the PV/FC/BG system, the absence of wind turbines increases the size and subsequent cost of components. Compared to PV/WT/FC/BG, there are approximately 211.80% and 192.28% increases in the costs of PVs and fuel cells, respectively. In the WT/FC/BG system, WT share half of the total NPC (50.29%) and BGs share 32.30% of the total net present cost in the WT/FC/BG system.

4.2.1. Impact of load increment on system sizing

Load Increment refers to the gradual increase in the demand for electrical power by consumers over a specified period. As load increases, the LPSP, which evaluates the likelihood of the system failing to meet

demand, may rise without appropriate system resizing. System resizing involves adjusting the capacity and configuration of energy generation and storage components, such as adding more PV modules, WT, BG, and FC to the HRES. This ensures a stable, efficient, and cost-effective power supply while minimizing COE and HHD, as the objective function states. Load increment is key to dynamic energy system planning, ensuring future demand is met with high reliability and minimal power interruptions.

In Fig. 12, the PV/WT/FC/BG arrangement exhibits substantial modifications to meet the heightened demand from the baseline load to a 50% load augmentation. The number of PV modules increase steadily from at 50% load increment, displaying a 187.6% increase. This escalation underscores a strategic reliance on solar energy as demand rises. Conversely, the number of WT, FC, and hydrogen storage components fluctuates with load increments, reflecting their varied contributions to system stability. For instance, WT installations vary but generally stabilize or decrease slightly, maintaining adaptability within the system. FC and hydrogen storage see notable increases, such as the FC count rising from 34 at baseline to 68 at 50% load, demonstrating their growing role in balancing energy supply amidst escalating demand. At a 10% load increment, the system sees moderate adjustments, with slight increases in number of PV modules from 827 to 1421 and reductions in number of other components like WT from 75 to 67, and FC from 34 to 47. This trend continues up to a 30% load increment, where optimization efforts appear effective in balancing energy generation and storage capacities. Beyond this point, particularly at 40% and 50% load increments, there's a noticeable increase in PV and WT installations, indicating a strategic shift towards maximizing renewable energy sources. Throughout these increments, the COE and the HHD remain within stable ranges, with COE varying between 0.1574 and 0.1634 \$/kWh and HHD between 0.1732 and 0.2719. This stability emphasizes the effectiveness of the NSGA-II algorithm in managing system performance and environmental impact across varying load scenarios.

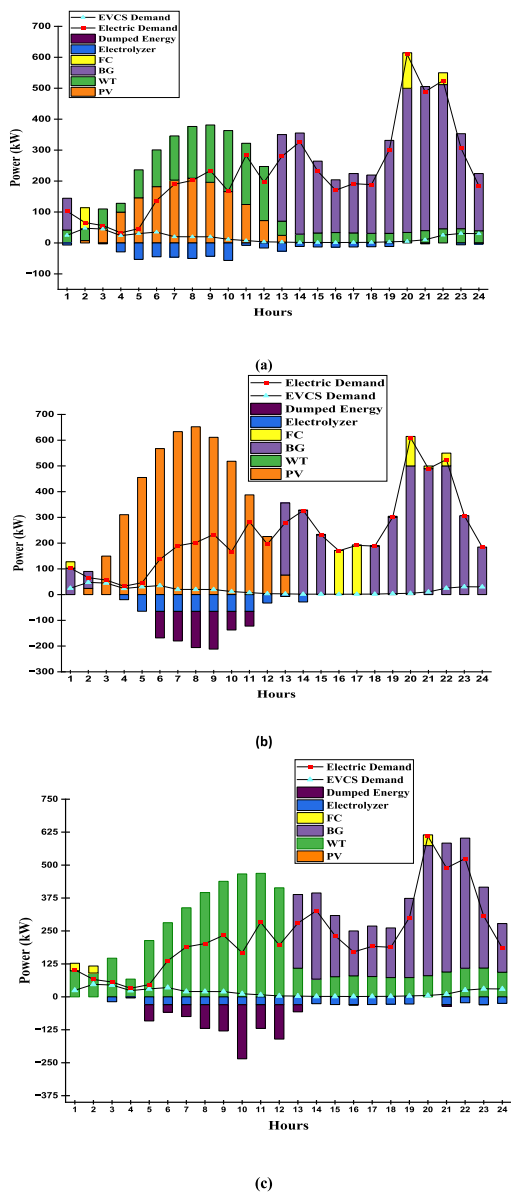


Fig. 9. Scenario of power dynamics for (a) PV/WT/FC/BG (b) PV/FC/BG, and WT/FC/BG-based hybrid systems on a regular day. Production and supply are indicated by the positive Y-axis, and storage and disposal of extra production are indicated by the negative axis orientation.

However, as load increments rise, there's a notable increase in HHD, signaling potential environmental concerns despite operational efficiencies. This highlights the ongoing challenge of optimizing HRES for both economic viability and environmental sustainability under varying load conditions.

Table 8
Data for Energy Efficiency on the optimal HRES configurations.

HRES Variant	Energy supplied (kWh/yr)	Total energy generation (kWh/yr)	Excess energy (kWh/yr)	Energy loss (kWh/yr)	System efficiency	Excess Energy Utilization Factor (EUF)
PV/WT/FC/BG	2,073,411	2,090,103	273,130	49,202	85%	82%
PV/FC/BG	2,478,717	2,498,561	808,788	143,641	62%	82%
WT/FC/BG	2,129,903	2,140,030	310,855	156,871	78%	50%

4.2.2. Impact of load increment on system cost

As the HRES adjusts to load increments from baseline to 50%, both the COE and the TNPC undergo significant changes, reflecting the system's dynamic response to increased energy demands. In Fig. 13, TNPC exhibits a consistent upward trend, driven by substantial expansions in PV modules, WT, FC, and hydrogen storage components. The quantity of PV modules rises from 827 at baseline to 2374 with a 50% load increase, whereas WT, FC, and hydrogen storage exhibit corresponding increases. This scaling requires higher initial investments and ongoing operational expenses, contributing to TNPC rises ranging from \$389,855 (9.77% increase) at a 10% load increment to \$1,859,151 (46.59% increase) at a

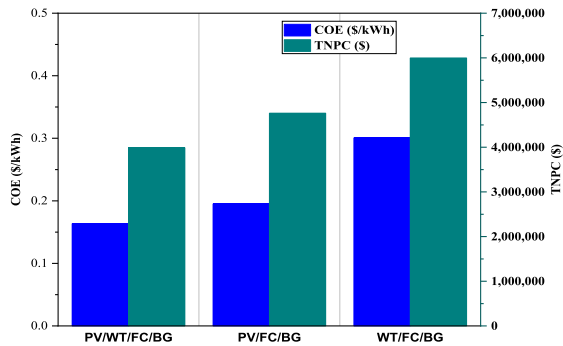


Fig. 10. NPC and COE for the three configurations.

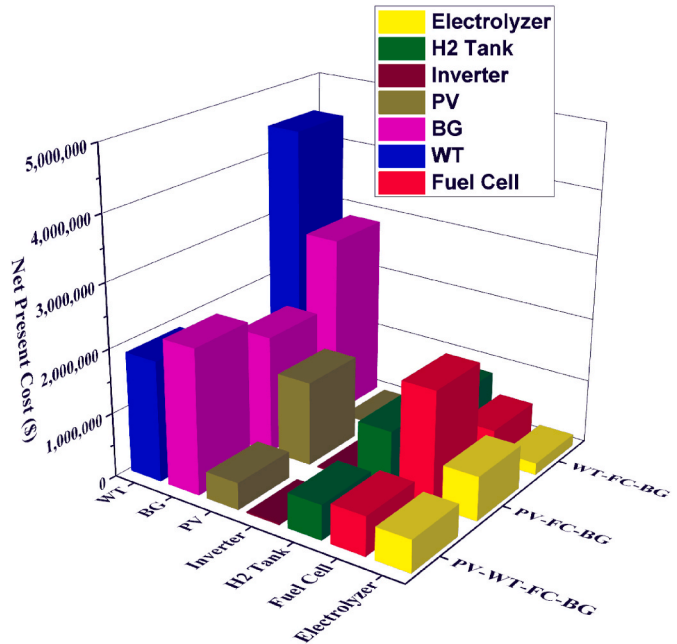


Fig. 11. Comparison of net present costs of different components in each configuration.

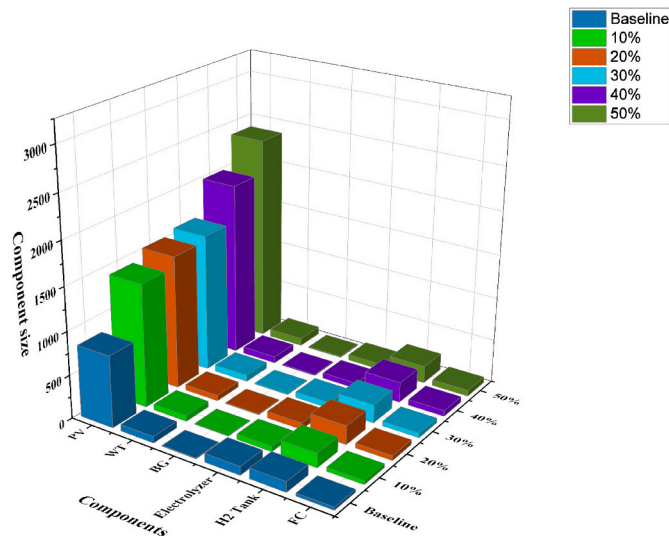


Fig. 12. Effect of load increment on system sizing of PV/WT/FC/BG.

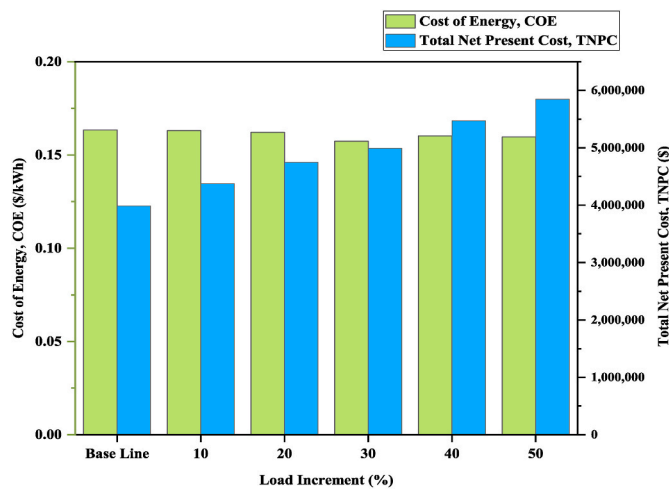


Fig. 13. Impact of load increment on system cost metrics.

50% load increment compared to the baseline value of \$3,988,169. The increase in TNPC is directly linked to the expansion of system components, each contributing to higher capital and operational costs.

In contrast, the COE maintains stability between 0.1574 \$/kWh and 0.1634 \$/kWh throughout the load increments. This stability underscores the efficiency of the NSGA-II optimization algorithm in managing energy production costs effectively despite system expansion. By maintaining a consistent COE, the system demonstrates its capability to control energy costs while scaling up to meet escalating demands, ensuring sustainable energy production over the long term.

4.3. Environment analysis

The aim of this study is to scrutinize the environmental effects of hybrid power systems, tracking the CO₂ equivalent emissions of each element throughout its full operational life. Moreover, this research also takes into account the influence of these contaminants on human well-being and the surrounding ecological environment. The data depicted in Fig. 14 shows the performance parameters in the three configurations. The PV/WT/FC/BG and the WT/FC/BG systems show almost similar behavior in terms of the parameters considered, with the WT/FC/BG system being slightly ahead. However, the PV/WT/FC/BG system is our configuration of interest due to its lowest COE and highest system

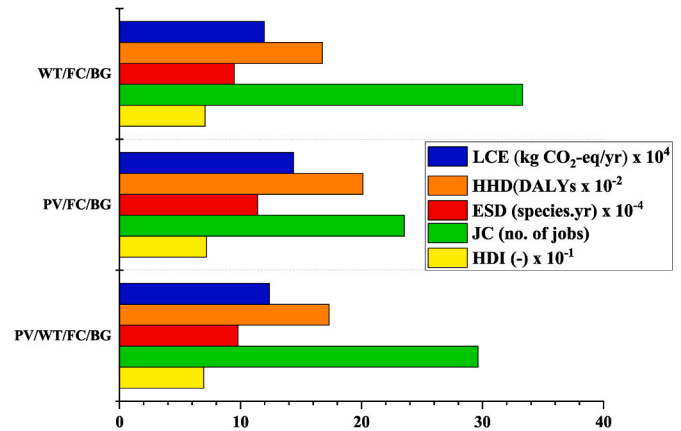


Fig. 14. Effect of various HRESs on human health, ecosystem, and social benefit factors.

efficiency. This system is expected to have the moderate detrimental impact on both human health and ecology (HHD and ESD) out of all three configurations. This system's annual aggregate of life cycle emission (LCE) amounts to 123,730 kg. It is evident from Fig. 15 that the majority of these emissions, approximately 69.90%, are attributed to BG, while wind turbines contribute the least amount, accounting for only 2.10% of the total.

In the PV/FC/BG system, the absence of wind turbines leads to a significant increase in PV size and excess energy, which results in higher utilization of FCs and lesser operational hours of BG. Hence, a subsequent reduction of 22.65% in BG emissions is observed compared to PV/WT/FC/BG (Fig. 15). However, the inclusion of more PV modules having a higher emission factor results in an approximate 20.02% rise in emissions. The presence of substantial hardware components, specifically PVs, amplifies the overall emission generated by a PV/FC/BG-based system.

4.3.1. Penalty for carbon emission

This study encompassed an examination of the environmental impact linked to the components of a HRES, with a specific focus on assessing CO₂ emissions throughout their lifecycle. It is to be noted that

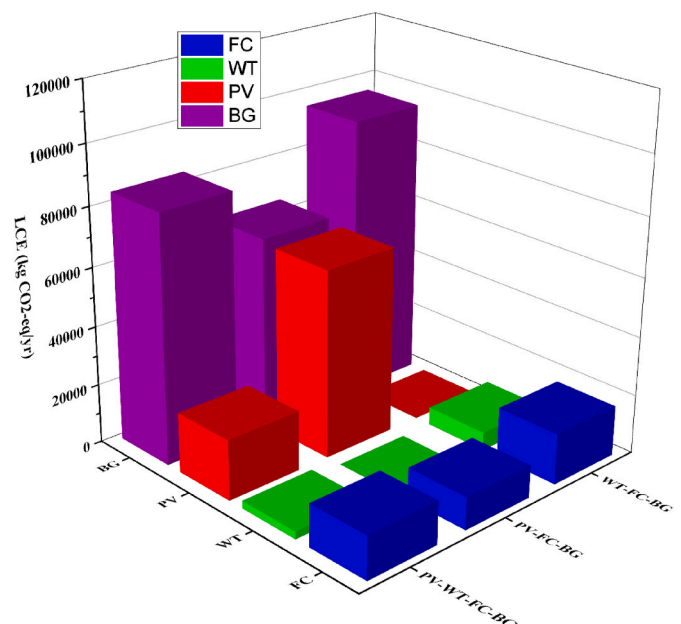


Fig. 15. LCE shares from various hybrid system components.

renewable energy sources do not emit any CO₂ during their operational phase. However, the HRES system is associated with a metric known as the LCE, which quantifies CO₂ emissions in kilograms per kilowatt-hour (kg CO₂-eq/kWh). This LCE incorporates the carbon emissions tied to the manufacturing, transportation, and recycling processes, representing the carbon footprint of renewable energy components within the HRES. The study's results revealed that the combined system of WT, FC, and BG produces the least CO₂-eq annually out of the three configurations, approximately 119,720 kg. Additionally, this study factored in a carbon emission tax of \$0.015 per kilogram of CO₂ emissions, as documented by Maisanam et al. [82]. The yearly cost stemming from the emission amounted to \$1795.80. The costs associated with carbon penalties for the PV/WT/FC/BG and the PV/FC/BG systems are \$1855.95 and \$2155.35, respectively.

Incorporating the carbon penalties, the computed electricity costs stand at: 0.1634 \$/kWh for the PV/WT/FC/BG setup, 0.1951 \$/kWh for the PV/FC/BG configuration, and 0.2457 \$/kWh for the WT/FC/BG arrangement within the HRES framework.

4.3.2. Fuel economy and CO₂ emission reduction

The research outcomes indicate the possibility of significant diesel consumption savings for household lighting, electrical demand, and EVCS demand in the specified region. By substituting the conventional diesel generator power plant of the study area with the recommended hybrid configuration, substantial benefits will be obtained. Given the recent energy crisis in Bangladesh, the use of diesel generators has surged, resulting in heightened CO₂ emissions. Each household and commercial center relies multiple light sources. For supplying the load equivalent to the proposed system, a diesel generator-based power plant would require around 220,858.68 gallons of diesel fuel when considered a diesel generator will use 0.4 L of diesel per kWh produced [83]. Considering that the CO₂ emission coefficient for diesel is 2.68 kg of CO₂ per liter [84], the implementation of the suggested PV/WT/BG/FC-based HRES system has the potential to reduce CO₂ emissions by approximately 2,116,860.42 kg equivalent to diesel fuel CO₂ emission annually. This reduction can prove highly effective in mitigating CO₂ levels and also lead to savings equivalent to the cost of diesel fuel required to meet the demand, thereby bolstering the economy in various other aspects.

4.4. Analysis of social welfare

This study estimates two significant social benefit components, namely the HDI and the JC, under optimum settings. The excess energy generated by the energy system has the potential to enhance the overall quality of life for individuals residing in the area under investigation. As indicated in Fig. 14, the projected HDI values for the configurations of PV/WT/FC/BG, PV/FC/BG, and WT/FC/BG stand at 0.6959, 0.7205, and 0.7067, respectively. The JC factor, which signifies employment creation potential, is highest for the WT/FC/BG configuration with approximately 34 jobs/MW, followed by PV/WT/FC/BG with 30 jobs/MW, and PV/FC/BG with 24 jobs/MW over a project lifetime of 25 years. Implementation of a hybrid renewable system for electricity purposes will lead to an improvement in the fundamental lighting infrastructure of the community, hence alleviating the occurrence of frequent load-shedding incidents. According to Sharma et al. [85] implementing such improvements can effectively increase the quality of children's home study hours, hence mitigating the negative effects of eye strain and other health-related issues. The provision of enhanced illumination options is expected to positively impact engagement in educational activities, hence potentially leading to an improvement in the existing literacy rates. The proposed system will contribute to the enhancement of the business environment within the community through the provision of a stable electrical supply, hence enabling longer night-time illumination hours.

4.5. Results validation by comparison

The optimal configurations of the HRES largely depend on the type of load profiles, technical and economic details of power generating systems, climatic data, energy management strategy, energy storage device, and the optimization techniques. Therefore, the direct comparative analysis between the HRES with the literature is unlikely possible. Nonetheless, this analysis offers some insight into the findings of this study in comparison to other research projects. Within the trio of optimized configurations, the PV/WT/FC/BG system is distinguished as the most economical, offering an energy cost of only 0.1634 \$/kWh and recording a net present cost of \$3,988,168.98. The financial and ecological advantages of this system are validated by recent scholarly publications. Das et al. [75] conducted a study in Bangladesh that reported a cost of 0.234 \$/kWh and an annual emission of 40,239 kg CO₂-eq. Mojumder et al. [86] optimized PV/WT/Diesel/Natural gas genset/Batt-based HRESs using HOMER in Bangladesh and results found that the COEs in the ranges of 0.169–0.266 \$/kWh. In their study, Salameh et al. [87] carried out an optimization of a system based on PV/FC/DG/FCES in the UAE climatic condition. The optimized system demonstrated an energy cost of 0.341 \$/kWh and an annual CO₂ emission of 3,844,051 kg. In their proposal, Xu et al. introduced an ideal PV/FC/WT system that yielded an energy cost of 0.226 \$/kWh [88]. As per the optimization investigation undertaken by Roy et al. [47] a system incorporating PV, WT, FC, and biomass generators demonstrated the lowest energy cost, amounting to 0.1967 \$/kWh in India. In another study, Yadav et al. [89] reported that the lowest COE of 0.189 \$/kWh can be achieved by the PV/WT/FC/Batt with electrolyzer option in Indian climatic condition. A PV/FC/Batt with hydrogen storage-based HRES offers a COE of 0.3159/kWh in Pakistan [90]. The approach described in this study reveals notable improvements in economic and environmental consequences, positioning it as a strong and competitive option.

5. Conclusions

This study presents a comprehensive hybrid power system designed for dual load operation, evaluating its economic feasibility, environmental impact, and societal benefits in Rajshahi. The MSW-based biogas with the solar PV and wind turbines along with the hydrogen storage options are optimized using NSGA-II. The several key parameters are also reported, which will provide a foundation for decision-makers in implementing such project in Bangladesh. The PV/WT/FC/BG system offers the highest energy efficiency of 84.58% with the least COE of 0.1634 \$/kWh and a net present cost of \$3,988,169. The HHD of this system is measured at about 0.1732 DALYs, which is lower than the PV/FC/BG system (0.2012 DALYs). The lifecycle emissions for the PV/WT/FC/BG system results in annual CO₂ emissions of 123,730 kg, with potential annual CO₂ emission reductions of about 2.1 million kg compared to the diesel only system. The BGs contribute 69.90% to CO₂ emissions, whereas WT contributes only 2.10%. The WT/FC/BG HRES system offers the highest number of job creation potential, generating 34 jobs per MW over 25 years. Following closely, the PV/WT/FC/BG system creates 30 jobs per MW over the same period. In terms of development index, the PV/FC/BG system ranks highest with a score of 0.7205, while the PV/WT/FC/BG system comes next with a score of 0.696. The WT/FC/BG model incurs an annual carbon penalty of \$1795.80, lower than the PV/WT/FC/BG and PV/FC/BG models, with costs of \$1855.95 and \$2155.35, respectively. The PV/WT/FC/BG system relies more heavily on photovoltaic (PV) modules as demand rises, with a 187.6% increase at a 50% load increment. With the changes of load demand, the COE varies in the ranges of 0.1574–0.1634 \$/kWh and the HHD varies between 0.1732 and 0.2719 for the PV/WT/FC/BG-based HRES.

The application of the NSGA-II algorithm for hybrid energy system optimization highlights several areas for future expansion. Integrating a

grid connection could reduce unmet electric loads to zero, improving reliability and efficiency. Selling surplus energy back to the grid would enhance economic benefits, increasing the system's cost-effectiveness. Additionally, expanding the multi-objective optimization framework to include more than two objective functions could provide more comprehensive solutions, although it would require further refinement of the NSGA-II algorithm to handle the increased complexity.

CRedit authorship contribution statement

Asif Jaman: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Rafid Al Mahmud:**

Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Barun K. Das:** Writing – review & editing, Supervision, Software, Methodology, Investigation, Conceptualization. **Mohammad Shahed H.K. Tushar:** Writing – review & editing, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A

Details on the technical and economic features of the proposed HRES

Component Name	Technical Specifications	Data	Economic Details	Data
PV module [91,92]	Model	SPR-E20-327	Initial expenditure	1300\$/kW
	Maximum power, R_{pv}	327W	O&M expense	20\$/kW/yr
	Solar efficiency, η_{pv}	20.4%	Restoration value	0
	Nominal voltage, V_{mpp}	54.7V		
	Nominal current, I_{mpp}	5.98A		
	Thermal coefficient of power, α_p	−0.35%/°C		
	Derating ratio, f_{pv}	88%		
Wind turbine [93,94]	Lifespan	25yrs		
	Maximum power, R_{WT}	5 kW	Initial expenditure	420\$/kW
	Rotor diameter, d	5.8 m	O&M expense	0.011\$/kWh
	Minimum wind speed, V_{cut-in}	1.8 m/s	Restoration value	0
	Nominal wind speed, V_{rated}	12 m/s		
	Maximum wind speed, $V_{cut-out}$	65 m/s		
	Rotor axis height	10 m		
Biogas Generator [95]	Number of blades	5 (H- Darrius type)		
	Lifespan	20 years		
	Maximum power	50 kW	Initial expenditure	1600\$/kW
	Lifespan	25,000 h	O&M expense	0.025\$/h
Inverter [96]			Restoration value	1280\$/kW
	Operating power, P_{INV}	1 kW	Initial expenditure	300\$/kw
	Conversion rate, η_{inv}	96%	O&M expense	0
	Lifespan	15 Years	Restoration value	300\$/kW
Fuel Cell [97]	Maximum power	2 kW	Initial expenditure	3000 \$/kW
	Conversion rate, η_{fc}	50%	O&M expense	0.02 × initial expenditure
	Lifespan	5 years	Restoration value	2500 \$/kW
Hydrogen Tank [97]	Maximum capacity	6 kg	Initial expenditure	660\$/kW
	Lifespan	20 years	O&M expense	0.02 × initial expenditure
Electrolyzer [97]	Maximum power	2 kW	Initial expenditure	2000 \$/kW
	Lifespan	15 years	O&M expense	20\$/unit
			Restoration value	1500 \$/kW

Nomenclature

S_{swept}	Total swept area (m ²)
C_{cap}	Capital cost (\$)
$C_{rep,comp}$	Final replacement price
$C_{cap,comp}$	Initial purchase price
C_{fuel}	Fuel cost (\$)
$C_{O\&M}$	Operation and maintenance cost
C_{rep}	Replacement cost
$C_{salvage}$	Salvage values
COE	Cost of energy
COH	Cost of hydrogen
E	Energy generation (kWh)
EEF	Excess energy fraction
ESD	Ecosystem damage
EUF	Energy utilization factor
EV	Electric vehicle
EVCS	Electric vehicle charging station

(continued on next page)

(continued)

$E_{\text{Electrolyzer}}$	Energy consumption by electrolyzer
E_{Fc}	Energy generation by fuel cell
E_{BG}	Energy generation by biogas (kWh)
E_{S}	Annual provided load
f_{pv}	Derating factor of PV (%)
F_{fuel}	Annual fuel consumption (L/year)
G_{C}	Per capita solid waste generation at present (kg/capita/day)
HHV_{H_2}	Higher heating value of hydrogen
H_{HS}	Total quantity of hydrogen stored in the storage (kg)
$H_{\text{electrolyzer}}$	Hydrogen generation in electrolyzer
hr	Hours of operation (hours)
HDI	Human development index
HES	Hybrid energy system
HHD	Human health damage
HRES	Hybrid renewable energy system
i	Annual interest rate (%)
I_{T}	Solar irradiation incident (kW/m^2)
I_{S}	Solar irradiation at the standard test conditions
JC	Job creation
L	Project lifetime (year)
LPSP	Loss of power supply probability
M	Annual cattle dung production
N_{comp}	Number of components
$N_{\text{rep.comp}}$	Measures how often a certain component needs to be replaced
P_{C}	Number of population at present
P_{in}	Input power
P_{out}	Output power
P_{WT}	Power generation by WT (kW)
Q_{br}	Recoverable biogas flow (m^3/y)
SWG_{F}	The annual solid waste generation in future (Ton per year)
T_{C}	PV cell temperature ($^{\circ}\text{C}$)
T_{S}	Cell temperature at standard condition ($^{\circ}\text{C}$)
VAWT	Vertical axis wind turbine
V_{b}	Volume of biogas generated from feedstock
$V_{\text{cut.in}}$	Cut-in wind velocity (m/s)
$V_{\text{cut.ou}}$	Cut-off wind velocity (m/s)
V_{rated}	Rated wind velocity (m/s)
W_{pv}	Power generation by PV (kW)
W_{rated}	Rated power (kW)
W_{RPV}	Rated power of the PV module
x_{f}	Annual increased rate of per capita waste generation in future.
Y_{f}	Annual increase rate of population growth in the future

References

- [1] World population review [cited 2024 20.06.2024]; Available from: <https://worldpopulationreview.com/countries/bangladesh>; 2024.
- [2] Zhao Q, et al. Conceptual design and optimization of integrating renewable energy sources with hydrogen energy storage capabilities. *Int J Hydrogen Energy* 2024;79: 1313–30. <https://doi.org/10.1016/j.ijhydene.2024.07.071>.
- [3] Karimi H. A tri-objectives scheduling model for renewable-hydrogen-based microgrid system considering hydrogen storage system and demand-side management. *Int J Hydrogen Energy* 2024;68:1412–22. <https://doi.org/10.1016/j.ijhydene.2024.04.330>.
- [4] Bangladesh power development board [cited 2024 02.10.2024]; Available from: <https://misc.bpdb.gov.bd/power-generation-unit>; 2024.
- [5] Gökçek M, et al. Optimum sizing of hybrid renewable power systems for on-site hydrogen refuelling stations: case studies from Türkiye and Spain. *Int J Hydrogen Energy* 2024;59:715–29. <https://doi.org/10.1016/j.ijhydene.2024.02.068>.
- [6] Hossain MB, et al. A power dispatch allocation strategy to produce green hydrogen in a grid-integrated offshore hybrid energy system. *Int J Hydrogen Energy* 2024;62: 1103–12. <https://doi.org/10.1016/j.ijhydene.2024.03.051>.
- [7] Khan ZH, Ahmed O. Assessment of noise pollution in Rajshahi city and remediation. *World J Adv Res Rev* 2023;18(2):346–51. <https://doi.org/10.30574/wjarr.2023.18.2.0830>.
- [8] Nouri A, Lachheb A, El Amraoui L. Optimizing efficiency of Vehicle-to-Grid system with intelligent management and ANN-PSO algorithm for battery electric vehicles. *Elec Power Syst Res* 2024;226:109936. <https://doi.org/10.1016/j.epsr.2023.109936>.
- [9] Basnet S, et al. Optimal integration of hybrid renewable energy systems for decarbonized urban electrification and hydrogen mobility. *Int J Hydrogen Energy* 2024;83:1448–62. <https://doi.org/10.1016/j.ijhydene.2024.08.054>.
- [10] Jaman A, et al. Techno-econo-environmental analysis of sustainable hybrid solar-wind-biogas using municipal solid waste-based grid independent power plant with dual mode energy storage strategy. *Energy* 2024;307:132777. <https://doi.org/10.1016/j.energy.2024.132777>.
- [11] Islam MR, et al. Site suitability assessment for solar power plants in Bangladesh: a GIS-based analytical hierarchy process (AHP) and multi-criteria decision analysis (MCDA) approach. *Renew Energy* 2024;220:119595. <https://doi.org/10.1016/j.renene.2023.119595>.
- [12] Molu RJJ, et al. A techno-economic perspective on efficient hybrid renewable energy solutions in Douala, Cameroon's grid-connected systems. *Sci Rep* 2024;14 (1):13590. <https://doi.org/10.1038/s41598-024-64427-4>.
- [13] Das BK, et al. Optimum design, socioenvironmental impact, and exergy analysis of a solar and rice husk-based off-grid hybrid renewable energy system. *Int Trans Electr Energy Syst* 2023;2023(1):3597840. <https://doi.org/10.1155/2023/3597840>.
- [14] Wang C, et al. Hybrid energy storage capacity configuration strategy for virtual power plants based on variable-ratio natural gas-hydrogen blending. *Int J Hydrogen Energy* 2024;58:433–45. <https://doi.org/10.1016/j.ijhydene.2024.01.175>.
- [15] Hussam WK, Abdul-Niby M, Sheard GJ. Techno-economic analysis and optimization of hydrogen production from renewable hybrid energy systems: shagaya renewable power plant-Kuwait. *Int J Hydrogen Energy* 2024;58:56–68. <https://doi.org/10.1016/j.ijhydene.2024.01.153>.
- [16] Das BK, et al. Techno-economic and environmental assessment of a hybrid renewable energy system using multi-objective genetic algorithm: a case study for remote Island in Bangladesh. *Energy Convers Manag* 2021;230:113823. <https://doi.org/10.1016/j.enconman.2020.113823>.
- [17] Molu RJJ, et al. Optimizing technical and economic aspects of off-grid hybrid renewable systems: a case study of manoka island, Cameroon. *IEEE Access* 2023; 11:130909–30. <https://doi.org/10.1109/ACCESS.2023.3332693>.
- [18] Alonso AM, et al. Techno-economic assessment on hybrid energy storage systems comprising hydrogen and batteries: a case study in Belgium. *Int J Hydrogen Energy* 2024;52:1124–35. <https://doi.org/10.1016/j.ijhydene.2023.06.282>.
- [19] Tang Y, et al. Energy management of electric-hydrogen hybrid energy storage systems in photovoltaic microgrids. *Int J Hydrogen Energy* 2024;80:1–10. <https://doi.org/10.1016/j.ijhydene.2024.07.017>.
- [20] Abdolmaleki L, Berardi U. Hybrid solar energy systems with hydrogen and electrical energy storage for a single house and a midrise apartment in North

- America. *Int J Hydrogen Energy* 2024;52:1381–94. <https://doi.org/10.1016/j.ijhydene.2023.11.222>.
- [21] Chamout MW, et al. Simulation and analysis of hybrid hydrogen-battery renewable energy storage for off-electric-grid Dutch household system. *Int J Hydrogen Energy* 2024;70:170–82. <https://doi.org/10.1016/j.ijhydene.2024.05.106>.
- [22] Le TT, et al. Fueling the future: a comprehensive review of hydrogen energy systems and their challenges. *Int J Hydrogen Energy* 2024;54:791–816. <https://doi.org/10.1016/j.ijhydene.2023.08.044>.
- [23] Abidin Z, Al Khafaf N, McGrath B. Feasibility of hydrogen hybrid energy systems for sustainable on- and off-grid integration: an Australian REZs case study. *Int J Hydrogen Energy* 2024;57:1197–207. <https://doi.org/10.1016/j.ijhydene.2024.01.122>.
- [24] Rathod AA, S B. Optimization of stand-alone hybrid renewable energy system based on techno-socio-enviro-financial perspective using improved red-tailed hawk algorithm. *Appl Energy* 2024;376:124137. <https://doi.org/10.1016/j.apenergy.2024.124137>.
- [25] Modu B, et al. Operational strategy and capacity optimization of standalone solar-wind-biomass-fuel cell energy system using hybrid LF-SSA algorithms. *Int J Hydrogen Energy* 2024;50:92–106. <https://doi.org/10.1016/j.ijhydene.2023.07.215>.
- [26] Ahmed MM, et al. Energy management and sizing of a stand-alone hybrid renewable energy system for community electricity, fresh water, and cooking gas demands of a remote island. *Energy Convers Manag* 2024;299:117865. <https://doi.org/10.1016/j.enconman.2023.117865>.
- [27] Yan C, et al. Effect of various design configurations and operating conditions for optimization of a wind/solar/hydrogen/fuel cell hybrid microgrid system by a bio-inspired algorithm. *Int J Hydrogen Energy* 2024;60:378–91. <https://doi.org/10.1016/j.ijhydene.2024.02.004>.
- [28] Tatti R, et al. Hydrogen storage integrated in off-grid power systems: a case study. *Int J Hydrogen Energy* 2024;79:164–76. <https://doi.org/10.1016/j.ijhydene.2024.06.308>.
- [29] Al Wahedi A, Bicer Y. Techno-economic optimization of novel stand-alone renewables-based electric vehicle charging stations in Qatar. *Energy* 2022;243:123008. <https://doi.org/10.1016/j.energy.2021.123008>.
- [30] Cuesta M, Castillo-Calzadilla T, Borges C. A critical analysis on hybrid renewable energy modeling tools: an emerging opportunity to include social indicators to optimise systems in small communities. *Renew Sustain Energy Rev* 2020;122:109691. <https://doi.org/10.1016/j.rser.2019.109691>.
- [31] Padrón I, et al. Assessment of hybrid renewable energy systems to supplied energy to autonomous desalination systems in two islands of the canary archipelago. *Renew Sustain Energy Rev* 2019;101:221–30. <https://doi.org/10.1016/j.rser.2018.11.009>.
- [32] Mayer MJ, Szilágyi A, Gróf G. Environmental and economic multi-objective optimization of a household level hybrid renewable energy system by genetic algorithm. *Appl Energy* 2020;269:115058. <https://doi.org/10.1016/j.apenergy.2020.115058>.
- [33] Daramola AS, et al. A cost-effective and ecological stochastic optimization for integration of distributed energy resources in energy networks considering vehicle-to-grid and combined heat and power technologies. *J Energy Storage* 2023;57:106203. <https://doi.org/10.1016/j.est.2022.106203>.
- [34] Yilmaz F, Ozturk M, Selbas R. Development and assessment of a newly developed renewable energy-based hybrid system with liquid hydrogen storage for sustainable development. *Int J Hydrogen Energy* 2024;56:406–17. <https://doi.org/10.1016/j.ijhydene.2023.12.208>.
- [35] Fabianek P, Madlener R. Techno-economic analysis and optimal sizing of hybrid PV-wind systems for hydrogen production by PEM electrolysis in California and Northern Germany. *Int J Hydrogen Energy* 2024;67:1157–72. <https://doi.org/10.1016/j.ijhydene.2023.11.196>.
- [36] Okonkwo PC, et al. A techno-economic analysis of renewable hybrid energy systems for hydrogen production at refueling stations. *Int J Hydrogen Energy* 2024;78:68–82. <https://doi.org/10.1016/j.ijhydene.2024.06.294>.
- [37] Tariq AH, et al. Techno-economic and composite performance assessment of fuel cell-based hybrid energy systems for green hydrogen production and heat recovery. *Int J Hydrogen Energy* 2024. <https://doi.org/10.1016/j.ijhydene.2024.04.018>.
- [38] Ghirardi E, et al. The role of hydrogen storage and electric vehicles in grid-isolated hybrid energy system with high penetration of renewable. *Energy Convers Manag* 2024;302:118154. <https://doi.org/10.1016/j.enconman.2024.118154>.
- [39] Lee S, et al. Integrating hybrid PV/wind-based electric vehicles charging stations with green hydrogen production in Kentucky through techno-economic assessment. *Int J Hydrogen Energy* 2024;71:345–56. <https://doi.org/10.1016/j.ijhydene.2024.05.053>.
- [40] Li Y, et al. Technical and economic analysis of a hybrid PV/wind energy system for hydrogen refueling stations. *Energy* 2024;303:131899. <https://doi.org/10.1016/j.energy.2024.131899>.
- [41] Hussam WK, et al. Techno-economic analysis and optimization of hydrogen production from renewable hybrid energy systems: shagaya renewable power plant-Kuwait. *Int J Hydrogen Energy* 2024;58:56–68. <https://doi.org/10.1016/j.ijhydene.2024.01.153>.
- [42] Allouhi A, Rehman S. Grid-connected hybrid renewable energy systems for supermarkets with electric vehicle charging platforms: optimization and sensitivity analyses. *Energy Rep* 2023;9:3305–18. <https://doi.org/10.1016/j.egy.2023.02.005>.
- [43] Al Afif R, Ayed Y, Maaitah ON. Feasibility and optimal sizing analysis of hybrid renewable energy systems: a case study of Al-Karak, Jordan. *Renew Energy* 2023;204:229–49. <https://doi.org/10.1016/j.renene.2022.12.109>.
- [44] Jia K, et al. Modeling and optimization of a hybrid renewable energy system integrated with gas turbine and energy storage. *Energy Convers Manag* 2023;279:116763. <https://doi.org/10.1016/j.enconman.2023.116763>.
- [45] Singh B, Kumar A. Optimal energy management and feasibility analysis of hybrid renewable energy sources with BESS and impact of electric vehicle load with demand response program. *Energy* 2023;278:127867. <https://doi.org/10.1016/j.energy.2023.127867>.
- [46] Praveenkumar S, et al. Techno-economic optimization of PV system for hydrogen production and electric vehicle charging stations under five different climatic conditions in India. *Int J Hydrogen Energy* 2022;47(90):38087–105. <https://doi.org/10.1016/j.ijhydene.2022.09.015>.
- [47] Roy D, Hassan R, Das BK. A hybrid renewable-based solution to electricity and freshwater problems in the off-grid Sundarbans region of India: optimum sizing and socio-enviro-economic evaluation. *J Clean Prod* 2022;372:133761. <https://doi.org/10.1016/j.jclepro.2022.133761>.
- [48] Sifakis N, Konidakis S, Tsoutsos T. Hybrid renewable energy system optimum design and smart dispatch for nearly Zero Energy Ports. *J Clean Prod* 2021;310:127397. <https://doi.org/10.1016/j.jclepro.2021.127397>.
- [49] Das BK, Tushar MSHK, Hassan R. Techno-economic optimisation of stand-alone hybrid renewable energy systems for concurrently meeting electric and heating demand. *Sustain Cities Soc* 2021;68:102763. <https://doi.org/10.1016/j.scs.2021.102763>.
- [50] Eisapour AH, Jafarpur K, Farjah E. Feasibility study of a smart hybrid renewable energy system to supply the electricity and heat demand of Eram Campus, Shiraz University; simulation, optimization, and sensitivity analysis. *Energy Convers Manag* 2021;248:114779. <https://doi.org/10.1016/j.enconman.2021.114779>.
- [51] Khan EU, et al. Techno-economic analysis of small scale biogas based polygeneration systems: Bangladesh case study. *Sustain Energy Technol Assessments* 2014;7:68–78. <https://doi.org/10.1016/j.seta.2014.03.004>.
- [52] Halder PK, et al. Feasibility analysis of implementing anaerobic digestion as a potential energy source in Bangladesh. *Renew Sustain Energy Rev* 2016;65:124–34. <https://doi.org/10.1016/j.rser.2016.06.094>.
- [53] Khan EU, Martin AR. Review of biogas digester technology in rural Bangladesh. *Renew Sustain Energy Rev* 2016;62:247–59.
- [54] El-houari H, et al. Feasibility evaluation of a hybrid renewable power generation system for sustainable electricity supply in a Moroccan remote site. *J Clean Prod* 2020;277:123534. <https://doi.org/10.1016/j.jclepro.2020.123534>.
- [55] K Hossain A, Badr O. Prospects of renewable energy utilisation for electricity generation in Bangladesh. *Renew Sustain Energy Rev* 2007;11(8):1617–49. <https://doi.org/10.1016/j.rser.2005.12.010>.
- [56] Abedin MA, Jahiruddin M. Waste generation and management in Bangladesh: an overview. *Asian J Medical Biol Res* 2015;1(1):114–20.
- [57] Tun MM, Juchelková D. Assessment of solid waste generation and greenhouse gas emission potential in Yangon city, Myanmar. *J Mater Cycles Waste Manag* 2018;20(3):1397–408. <https://doi.org/10.1007/s10163-017-0697-y>.
- [58] Halder PK, et al. Energy scarcity and potential of renewable energy in Bangladesh. *Renew Sustain Energy Rev* 2015;51:1636–49. <https://doi.org/10.1016/j.rser.2015.07.069>.
- [59] Eriksson O. Environmental technology assessment of natural gas compared to biogas. *Nat Gas* 2010;1:127–47.
- [60] Koholé YW, et al. An effective sizing and sensitivity analysis of a hybrid renewable energy system for household, multi-media and rural healthcare centres power supply: a case study of Kaele, Cameroon. *Int J Hydrogen Energy* 2024;49:1321–59. <https://doi.org/10.1016/j.ijhydene.2023.09.093>.
- [61] Mandal S, Das BK, Hoque N. Optimum sizing of a stand-alone hybrid energy system for rural electrification in Bangladesh. *J Clean Prod* 2018;200:12–27. <https://doi.org/10.1016/j.jclepro.2018.07.257>.
- [62] Chedid R, Akiki H, Rahman S. A decision support technique for the design of hybrid solar-wind power systems. *IEEE Trans Energy Convers* 1998;13(1):76–83. <https://doi.org/10.1109/60.658207>.
- [63] Anoune K, et al. Sizing methods and optimization techniques for PV-wind based hybrid renewable energy system: a review. *Renew Sustain Energy Rev* 2018;93:652–73.
- [64] Barragán-Escandón A, et al. Assessment of power generation using biogas from landfills in an equatorial tropical context. *Sustainability* 2020;12(7):2669.
- [65] Ceylan C, Devrim Y. Design and simulation of the PV/PEM fuel cell based hybrid energy system using MATLAB/Simulink for greenhouse application. *Int J Hydrogen Energy* 2021;46(42):22092–106. <https://doi.org/10.1016/j.ijhydene.2021.04.034>.
- [66] Kaabeche A, Bakelli Y. Renewable hybrid system size optimization considering various electrochemical energy storage technologies. *Energy Convers Manag* 2019;193:162–75. <https://doi.org/10.1016/j.enconman.2019.04.064>.
- [67] Hassan R, Das BK, Hasan M. Integrated off-grid hybrid renewable energy system optimization based on economic, environmental, and social indicators for sustainable development. *Energy* 2022;250:123823. <https://doi.org/10.1016/j.energy.2022.123823>.
- [68] Das BK, et al. Feasibility and techno-economic analysis of stand-alone and grid-connected PV/Wind/Diesel/Batt hybrid energy system: a case study. *Energy Strategy Rev* 2021;37:100673. <https://doi.org/10.1016/j.esr.2021.100673>.
- [69] Das BK, Al-Abdeli YM, Kothapalli G. Optimisation of stand-alone hybrid energy systems supplemented by combustion-based prime movers. *Appl Energy* 2017;196:18–33. <https://doi.org/10.1016/j.apenergy.2017.03.119>.
- [70] Shi Y, et al. Localized regional life cycle model research for the impacts of carbon dioxide on human health and ecosystem. *Sustain Prod Consum* 2022;29:36–45. <https://doi.org/10.1016/j.spc.2021.09.019>.

- [71] Huijbregts MAJ, et al. ReCiPe2016: a harmonised life cycle impact assessment method at midpoint and endpoint level. *Int J Life Cycle Assess* 2017;22(2):138–47. <https://doi.org/10.1007/s11367-016-1246-y>.
- [72] Sunny MR, et al. Techno-economic feasibility of stand-alone hybrid energy system with battery storage in educational buildings: a case study of Uttara University. *Energy Build* 2024;304:113852. <https://doi.org/10.1016/j.enbuild.2023.113852>.
- [73] Maqbool U, et al. Optimization of the renewable-energy-based micro-grid for rural electrification in northern region of India. *Clean Technol Environ Policy* 2020;22(3):579–90. <https://doi.org/10.1007/s10098-019-01801-1>.
- [74] Sawle Y, Gupta SC, Bohre AK. Socio-techno-economic design of hybrid renewable energy system using optimization techniques. *Renew Energy* 2018;119:459–72. <https://doi.org/10.1016/j.renene.2017.11.058>.
- [75] Das P, et al. Evaluating the prospect of utilizing excess energy and creating employments from a hybrid energy system meeting electricity and freshwater demands using multi-objective evolutionary algorithms. *Energy* 2022;238:121860. <https://doi.org/10.1016/j.energy.2021.121860>.
- [76] Dufo-López R, Cristóbal-Monreal IR, Yusta JM. Optimisation of PV-wind-diesel-battery stand-alone systems to minimise cost and maximise human development index and job creation. *Renew Energy* 2016;94:280–93. <https://doi.org/10.1016/j.renene.2016.03.065>.
- [77] Samy MM, Mosaad MI, Barakat S. Optimal economic study of hybrid PV-wind-fuel cell system integrated to unreliable electric utility using hybrid search optimization technique. *Int J Hydrogen Energy* 2021;46(20):11217–31. <https://doi.org/10.1016/j.ijhydene.2020.07.258>.
- [78] Verma S, Pant M, Snael V. A comprehensive review on NSGA-II for multi-objective combinatorial optimization problems. *IEEE Access* 2021;9:57757–91. <https://doi.org/10.1109/ACCESS.2021.3070634>.
- [79] Erdinc O, Uzunoglu M. Optimum design of hybrid renewable energy systems: overview of different approaches. *Renew Sustain Energy Rev* 2012;16(3):1412–25. <https://doi.org/10.1016/j.rser.2011.11.011>.
- [80] Deb K, et al. A fast and elitist multiobjective genetic algorithm: nsga-II. *IEEE Trans Evol Comput* 2002;6(2):182–97. <https://doi.org/10.1109/4235.996017>.
- [81] Biswas PP, et al. Multiobjective economic-environmental power dispatch with stochastic wind-solar-small hydro power. *Energy* 2018;150:1039–57. <https://doi.org/10.1016/j.energy.2018.03.002>.
- [82] Maisanam AKS, Biswas A, Sharma KK. Integrated socio-environmental and techno-economic factors for designing and sizing of a sustainable hybrid renewable energy system. *Energy Convers Manag* 2021;247:114709. <https://doi.org/10.1016/j.enconman.2021.114709>.
- [83] Diesel generator. 15/12/2024, https://energyeducation.ca/encyclopedia/Diesel_generator; 2023.
- [84] Michelin. How to calculate your fleet's carbon footprint [cited 2024 01.01.2024]; Available from: <https://connectedfleet.michelin.com/blog/calculate-co2-emissions/>; 2023.
- [85] Sharma R, et al. Do solar study lamps help children study at night? Evidence from rural India. *Energy Sustain Dev* 2019;50:109–16. <https://doi.org/10.1016/j.esd.2019.03.005>.
- [86] Mojumder MFH, et al. Techno-economic and environmental analysis of hybrid energy systems for remote areas: a sustainable case study in Bangladesh. *Energy Convers Manag* 2024;23:100664. <https://doi.org/10.1016/j.ecmx.2024.100664>.
- [87] Salameh T, et al. Integrated standalone hybrid solar PV, fuel cell and diesel generator power system for battery or supercapacitor storage systems in Khorfakkan, United Arab Emirates. *Int J Hydrogen Energy* 2021;46(8):6014–27. <https://doi.org/10.1016/j.ijhydene.2020.08.153>.
- [88] Xu C, et al. Data-driven configuration optimization of an off-grid wind/PV/hydrogen system based on modified NSGA-II and CRITIC-TOPSIS. *Energy Convers Manag* 2020;215:112892. <https://doi.org/10.1016/j.enconman.2020.112892>.
- [89] Yadav S, Kumar P, Kumar A. Techno-economic assessment of hybrid renewable energy system with multi energy storage system using HOMER. *Energy* 2024;297:131231. <https://doi.org/10.1016/j.energy.2024.131231>.
- [90] Tariq AH. Optimal sizing of photovoltaic/fuel cell-based energy system with autonomous oxygen production for hospitals in four climatic zones of Pakistan: an economic-energy-environmental feasibility analysis. *Renew Energy Focus* 2023;47:100494. <https://doi.org/10.1016/j.ref.2023.100494>.
- [91] Cano A, Arévalo P, Jurado F. Energy analysis and techno-economic assessment of a hybrid PV/HKT/BAT system using biomass gasifier: cuenca-Ecuador case study. *Energy* 2020;202:117727. <https://doi.org/10.1016/j.energy.2020.117727>.
- [92] PV module [cited 2023 15.12.2023]; Available from: <https://www.renewenergy.com.au/wp-content/uploads/Sunpower-E-Series-Residential-Solar-Panels-E20-327.pdf>; 2024.
- [93] Chong WT, et al. Techno-economic analysis of a wind-solar hybrid renewable energy system with rainwater collection feature for urban high-rise application. *Appl Energy* 2011;88(11):4067–77. <https://doi.org/10.1016/j.apenergy.2011.04.042>.
- [94] Vertical Axis wind turbine [cited 2024 05.01.2024]; Available from: <https://ellenenergy.en.made-in-china.com/>; 2024.
- [95] Yingjian L, et al. Energy balance and efficiency analysis for power generation in internal combustion engine sets using biogas. *Sustain Energy Technol Assessments* 2014;6:25–33. <https://doi.org/10.1016/j.seta.2014.01.003>.
- [96] Das BK, Tushar MSHK, Zaman F. Techno-economic feasibility and size optimisation of an off-grid hybrid system for supplying electricity and thermal loads. *Energy* 2021;215:119141. <https://doi.org/10.1016/j.energy.2020.119141>.
- [97] Jamshidi M, Askarzadeh A. Techno-economic analysis and size optimization of an off-grid hybrid photovoltaic, fuel cell and diesel generator system. *Sustain Cities Soc* 2019;44:310–20. <https://doi.org/10.1016/j.scs.2018.10.021>.